

INTEGRATION OF CLINICAL MACHINE LEARNING, NEUROIMAGING AND LLM FOR ALZHEIMER'S DISEASE DECISION SUPPORT SYSTEM

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled **INTEGRATION OF CLINICAL MACHINE LEARNING, NEUROIMAGING AND LLM FOR ALZHEIMER'S DISEASE DECISION SUPPORT SYSTEM** is the bonafied work of **MOHAMED ARSHAD J (RRN: 240282601050)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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VIVA-VOCE EXAMINATION

The viva-voce examination of the project work titled **“INTEGRATION OF CLINICAL MACHINE LEARNING, NEUROIMAGING AND LLM FOR ALZHEIMER’S DISEASE DECISION SUPPORT SYSTEM”** submitted by **MOHAMED ARSHAD J (RRN:240282601050)** is held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

The increasing availability of clinical and medical imaging data has created new opportunities for developing intelligent systems to support early diagnosis of Alzheimer's disease. However, manual analysis of clinical records and MRI scans is complex, time-consuming, and prone to errors. To address this challenge, this project proposes a Hybrid Multi-Modal Clinical Decision Support System that integrates machine learning, deep learning, and explainable artificial intelligence for accurate and interpretable diagnosis. The system utilizes clinical data such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) score, behavioral indicators, and medical history, along with MRI brain images for comprehensive analysis. Data preprocessing techniques including cleaning, normalization, and feature extraction are applied to ensure data quality and consistency.

A Random Forest model is employed to predict the likelihood of Alzheimer's disease based on clinical parameters, while a ResNet-18 convolutional neural network is used to classify disease stages from MRI images. To enhance interpretability, the system incorporates a large language model (LLM) that generates human-readable explanations for both prediction and classification results, along with personalized healthcare recommendations. The system follows a modular architecture with an intuitive user interface and efficient backend processing, enabling seamless interaction and real-time analysis. Experimental results demonstrate that the proposed system achieves high accuracy and reliability in Alzheimer's disease detection and stage classification.

By combining multi-modal data analysis with explainable AI, the system improves diagnostic efficiency and supports informed clinical decision-making. The proposed solution provides a scalable foundation for future advancements in AI-driven healthcare system

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
CDSS	Clinical Decision Support System
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
RF	Random Forest
MRI	Magnetic Resonance Imaging
API	Application programming Interface
CSV	Comma Separated Values
UI	User Interface
LLM	Large Language Model
UAT	User Acceptance Testing

CHAPTER 1

INTRODUCTION

1.1 GENERAL

In the modern healthcare ecosystem, the early detection and accurate diagnosis of neurodegenerative disorders such as Alzheimer's disease have become increasingly critical due to their progressive and irreversible nature. Alzheimer's disease is characterized by gradual cognitive decline, memory impairment, and structural changes in the brain, which often remain undetected in the early stages due to the subtlety of symptoms. Traditional diagnostic approaches rely heavily on manual clinical assessments and interpretation of neuroimaging data, which are often subjective, time-consuming, and prone to inconsistencies. This lack of integrated and efficient diagnostic mechanisms leads to delays in identifying the disease, ultimately affecting treatment outcomes and patient quality of life.

In recent years, the advancement of artificial intelligence has introduced new possibilities in the field of medical diagnosis, particularly in analyzing large-scale clinical and imaging datasets. However, existing AI-based solutions for Alzheimer's diagnosis are largely limited to single-modal approaches, focusing either on structured clinical data or on medical imaging such as MRI scans. These isolated systems fail to capture the complete diagnostic perspective, as Alzheimer's disease progression is influenced by both clinical indicators and structural brain abnormalities. Furthermore, many of these AI models function as black-box systems, providing predictions without clear explanations, which reduces trust and adoption among healthcare professionals.

To address these limitations, the proposed project introduces a Hybrid Multi-Modal Learning Framework for Alzheimer's Diagnosis that integrates clinical prediction, MRI-based classification, and explainable artificial intelligence into a unified system. The system is designed as an AI-powered Clinical Decision Support System (CDSS) capable of analyzing diverse data types and delivering accurate, transparent, and real-time diagnostic insights.

By combining machine learning techniques for clinical data analysis with deep learning models for neuroimaging and large language models for explainability, the system ensures a comprehensive and interpretable diagnostic process.

The proposed solution operates through a structured workflow where patient clinical data, including age, body mass index (BMI), Mini-Mental State Examination (MMSE) scores, behavioral indicators, and medical history, are first processed using a Random Forest-based predictive model to assess the risk of cognitive impairment. Upon detecting potential risk, the system proceeds to analyze MRI brain images using a ResNet-18 convolutional neural network, which classifies the disease into different stages such as no impairment, mild impairment, moderate impairment, and severe impairment. This multi-stage approach enhances diagnostic accuracy by validating clinical predictions with imaging-based evidence.

In addition to prediction and classification, the system incorporates an explainable AI component that generates human-readable interpretations of the results. This feature plays a crucial role in bridging the gap between AI-driven decisions and clinical understanding, allowing healthcare professionals to comprehend the reasoning behind each prediction. The integration of explainability ensures transparency, improves trust in the system, and supports informed decision-making in clinical environments.

The system is implemented as an interactive desktop application using Python and Tkinter, providing a user-friendly interface for healthcare professionals to input patient data, upload MRI images, and visualize diagnostic outputs in real time. The platform emphasizes usability, efficiency, and accessibility, ensuring that complex AI functionalities are delivered through a simple and intuitive interface. Additionally, the system maintains structured data management for clinical records, imaging datasets, and model outputs, enabling seamless processing and retrieval of information.

By replacing fragmented diagnostic processes with an integrated, AI-driven framework, the proposed system significantly improves the speed, accuracy, and reliability of Alzheimer's disease diagnosis. It reduces dependency on

manual interpretation, minimizes diagnostic errors, and enhances early detection capabilities. The system is particularly beneficial in clinical settings where timely decision-making is critical, as it provides rapid and consistent diagnostic support.

In conclusion, the proposed Hybrid Multi-Modal Clinical Decision Support System represents a significant advancement in the application of artificial intelligence in healthcare. By combining clinical data analysis, MRI-based deep learning, and explainable AI within a unified framework, the system delivers a scalable, efficient, and transparent solution for Alzheimer's diagnosis. This approach not only enhances diagnostic accuracy but also contributes to the development of intelligent, interpretable, and patient-centric healthcare systems, ultimately supporting improved treatment outcomes and better disease management.

1.2 EXISTING SYSTEM

In the current healthcare landscape, the diagnosis and management of Alzheimer's disease primarily rely on conventional clinical evaluation methods and standalone technological systems that operate in isolation. Most diagnostic procedures are based on cognitive assessments, patient history analysis, and manual interpretation of neuroimaging data such as MRI scans. These approaches, although widely practiced, are often time-consuming, subjective, and dependent on the expertise of medical professionals. As a result, the early detection of Alzheimer's disease becomes challenging, particularly in cases where symptoms are subtle and difficult to quantify using standard clinical techniques.

Existing systems for Alzheimer's diagnosis can broadly be categorized into two types: clinical data-based prediction systems and MRI-based image classification systems. Clinical prediction systems utilize structured patient data such as age, cognitive test scores, and medical history to estimate the likelihood of cognitive impairment. These systems commonly employ traditional machine learning algorithms such as Decision Trees, Support Vector Machines, and Random Forest models. While these methods provide

reasonable predictive performance, they are limited in their ability to capture the complex structural changes occurring in the brain, as they rely solely on tabular data without incorporating imaging insights.

On the other hand, MRI-based diagnostic systems focus on analyzing brain imaging data using deep learning techniques, particularly Convolutional Neural Networks (CNNs). These systems are effective in identifying structural abnormalities associated with Alzheimer's disease and are capable of classifying different stages of disease progression. However, such systems operate independently of clinical data and therefore lack a holistic understanding of the patient's condition. Additionally, most deep learning models function as black-box systems, providing predictions without clear explanations, which reduces their interpretability and limits their acceptance in clinical environments.

Another major limitation of existing systems is the lack of integration between clinical data analysis and medical image processing. Most healthcare institutions utilize separate tools or platforms for managing patient records and analyzing imaging data, resulting in fragmented workflows and inefficiencies in the diagnostic process. This disjointed approach increases the dependency on manual coordination between different diagnostic procedures, leading to delays in decision-making and reduced diagnostic accuracy.

Furthermore, existing AI-based healthcare systems often lack explainability mechanisms, making it difficult for healthcare professionals to understand the reasoning behind model predictions. The absence of transparency in decision-making reduces trust in AI systems and poses challenges in critical medical scenarios where interpretability is essential. In addition, many systems do not provide real-time diagnostic support or interactive interfaces, limiting their usability in practical clinical settings.

Existing solutions also face challenges related to data management, scalability, and accessibility. Clinical datasets and MRI imaging data are often stored in separate repositories without efficient integration, making it difficult to perform unified analysis. Moreover, many systems are not designed to handle

multi-modal data processing, which is essential for comprehensive disease diagnosis. The lack of centralized frameworks further complicates the process of maintaining consistency and reliability in diagnostic outputs.

1.2.1 Literature Survey

A literature survey is a fundamental component of any research work, as it provides a comprehensive understanding of existing methodologies, technologies, and research contributions related to the problem domain. It helps in identifying research gaps, evaluating existing solutions, and establishing the necessity for the proposed system. In this project, the focus is on developing a hybrid multi-modal framework for Alzheimer's disease diagnosis by integrating clinical data analysis, MRI-based deep learning, and explainable artificial intelligence. Several research works have been analyzed to understand advancements in machine learning, deep learning, and multimodal healthcare systems for Alzheimer's detection.

Recent advancements in Alzheimer's disease diagnosis have emphasized the integration of explainable artificial intelligence with multimodal data processing. In 2025, Sobhona Jahan et al. proposed a federated explainable AI-based system that integrates multimodal data sources, including clinical and psychological parameters, for Alzheimer's prediction. The study highlights the importance of privacy-preserving mechanisms and transparent decision-making in healthcare systems. However, the approach requires distributed infrastructure and is computationally intensive, limiting its scalability in real-world clinical environments.

In 2024, multiple studies contributed to improving hybrid learning approaches for Alzheimer's detection. Senthil Kumar et al. introduced a hybrid machine learning framework combining Convolutional Neural Networks (CNN) with Support Vector Machines (SVM) for early-stage prediction. This approach enhanced feature extraction and classification performance but was constrained by limited dataset size and generalization capability. Similarly, Gargi Pant Shukla et al. proposed a hybrid model integrating CNN, Random Forest, and XGBoost algorithms for MRI-based classification, achieving

improved accuracy in detecting Alzheimer's stages. However, the model's performance was highly dependent on preprocessing quality. Additionally, Rahmat Ullah et al. explored a novel approach using electromagnetic radar data combined with machine learning and autoencoder-based feature extraction for early detection. While this method introduced non-invasive diagnostic capabilities, it lacked extensive real-world clinical validation.

In 2023, research efforts focused significantly on machine learning-based clinical prediction models. Kavitha et al. developed a classification framework utilizing algorithms such as Support Vector Machine (SVM), Random Forest (RF), Decision Tree, and Gradient Boosting Machine (GBM) for early-stage Alzheimer's prediction using structured datasets like OASIS. The study demonstrated moderate accuracy levels but lacked multimodal integration. Similarly, Sudharsan et al. proposed a model combining Principal Component Analysis (PCA) for feature selection with SVM and RELM classifiers, improving classification efficiency through dimensionality reduction. However, the model was sensitive to feature selection and relied on limited training data. In addition, review-based studies conducted in 2023 analyzed the effectiveness of machine learning and deep learning techniques in Alzheimer's diagnosis, emphasizing the need for integrated and scalable solutions rather than isolated models.

In 2022, deep learning-based approaches gained prominence for MRI image analysis. Lee et al. proposed a 3D Convolutional Neural Network model combined with transfer learning for MRI-based Alzheimer's classification. The model achieved high diagnostic accuracy by effectively capturing structural brain patterns associated with the disease. However, it required large labeled datasets and high computational resources. Similarly, Liu et al. introduced a multimodal deep learning framework integrating MRI, PET, and clinical data to improve diagnostic performance through feature fusion. Despite achieving strong results, the approach faced challenges related to high data acquisition costs and limited scalability in clinical applications. Furthermore, review studies conducted during this period analyzed CNN-based MRI classification

techniques, highlighting their effectiveness while also pointing out the lack of experimental validation in some cases.

From the literature review, it is evident that most existing research works focus on either clinical data-based prediction or MRI-based classification independently. Although some multimodal approaches have been proposed, they often lack explainability, scalability, or practical implementation frameworks. Moreover, many systems operate as black-box models, limiting their adoption in real-world clinical environments due to the absence of transparent decision-making mechanisms.

Therefore, there is a clear research gap in developing a unified system that integrates clinical prediction, MRI-based deep learning, and explainable AI within a single framework. The proposed project aims to address this gap by designing a hybrid multi-modal Clinical Decision Support System that combines machine learning, deep learning, and explainability to provide accurate, reliable, and interpretable Alzheimer's disease diagnosis.

1.2.2 Limitations of Existing System

Despite significant advancements in artificial intelligence and medical data analysis, the existing systems for Alzheimer's disease diagnosis are not fully equipped to provide accurate, integrated, and real-time diagnostic support. Current approaches largely operate in isolation, focusing either on clinical data analysis or MRI-based imaging, without establishing a unified framework. These limitations affect both healthcare professionals and patients by reducing diagnostic accuracy, delaying early detection, and limiting interpretability of results. The major disadvantages of the existing systems are as follows:

❖ Lack of Multi-Modal Integration:

Most existing systems rely on a single type of data, either structured clinical data or MRI imaging, which results in incomplete analysis. The absence of integration between clinical indicators and neuroimaging reduces the overall effectiveness of diagnosis.

❖ **Limited Early Stage Detection:**

Early-stage Alzheimer's symptoms are subtle and difficult to quantify. Existing systems struggle to accurately identify mild cognitive impairment due to insufficient correlation between multiple data sources.

❖ **Dependence on Manual Interpretation:**

Traditional diagnostic methods rely heavily on clinicians to interpret cognitive assessments and MRI scans. This increases the chances of subjective bias and inconsistency in diagnosis.

❖ **Black-Box Nature of AI Models:**

Many machine learning and deep learning models used in healthcare operate as black-box systems, providing predictions without clear explanations. This lack of transparency reduces trust and usability in clinical environment

❖ **Fragmented Diagnostic Workflow:**

Clinical data analysis and MRI-based diagnosis are typically handled using separate systems, leading to inefficient workflows and delayed decision-making in healthcare settings.

❖ **Limited Explainability and Interpretability:**

Existing systems do not provide sufficient reasoning behind predictions, making it difficult for healthcare professionals to understand the contributing factors influencing the results.

❖ **High Dependency on Data Quality:**

The performance of current models is highly dependent on the quality of input data, including preprocessing of MRI images and accuracy of clinical datasets, which can lead to inconsistent results.

❖ **Scalability and Implementation Challenges:**

Many advanced models require high computational resources and large datasets, making them difficult to deploy in real-time clinical environments.

❖ **Lack of Real-Time Decision Support:**

Most systems are not designed to provide instant diagnostic outputs, which limits their usability in time-critical clinical scenarios.

❖ **Inefficient Data Management and Integration:**

Clinical datasets and imaging data are often stored separately without a unified framework, leading to difficulties in data synchronization and analysis.

1.3 PROPOSED SYSTEM

The proposed system is an AI-powered Hybrid Multi-Modal Clinical Decision Support System designed to address the limitations of existing Alzheimer's disease diagnostic approaches. The system integrates clinical data analysis, MRI-based deep learning, and explainable artificial intelligence into a unified framework, enabling accurate, efficient, and transparent diagnosis. It aims to modernize the diagnostic process by combining multiple sources of medical data and providing real-time, interpretable insights to healthcare professionals.

It enables clinicians to perform early-stage risk assessment using structured patient data such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) scores, behavioral indicators, and medical history. This data is processed using a machine learning model based on the Random Forest algorithm, which evaluates the likelihood of cognitive impairment. In cases where potential risk is identified, the system proceeds to analyze MRI brain images using a deep learning model based on the ResNet-18 convolutional neural network. This model classifies the patient's condition into different stages, including no impairment, mild impairment, moderate impairment, and severe impairment. By combining clinical prediction with imaging-based validation, the system ensures a comprehensive and reliable diagnostic process.

The system is implemented as an interactive desktop application developed using Python and Tkinter, providing a user-friendly interface for healthcare professionals. The interface allows users to input patient clinical data, upload MRI images, and view diagnostic results in real time. The design focuses on simplicity and accessibility, ensuring that complex AI functionalities are presented in an intuitive manner. The system also includes output panels that display prediction results, classification stages, and corresponding

recommendations, enabling quick and informed decision-making in clinical environments.

A key feature of the proposed system is the integration of explainable artificial intelligence through a large language model. This component generates human-readable explanations for both clinical predictions and MRI classification results. It provides insights into the factors influencing the diagnosis and offers stage-specific recommendations for treatment and care. This enhances transparency, builds trust among healthcare professionals, and supports better clinical decision-making by bridging the gap between AI predictions and human understanding.

The backend of the system is designed to manage multiple data types efficiently, including structured clinical datasets, MRI imaging data, model parameters, and prediction outputs. Data preprocessing mechanisms ensure that clinical inputs are transformed into suitable formats for machine learning models, while MRI images undergo normalization, resizing, and tensor conversion for compatibility with deep learning architectures. The system maintains organized data storage to enable efficient retrieval, processing, and analysis of information.

Security, reliability, and scalability are important considerations in the proposed system design. The modular architecture allows independent functioning of clinical prediction, MRI classification, and explainability modules, making the system easy to maintain and extend. The system is capable of handling large datasets and can be further enhanced to support cloud-based deployment, real-time monitoring, and integration with hospital information systems.

By replacing fragmented and single-modal diagnostic approaches with an integrated AI-driven framework, the proposed system significantly improves the accuracy and efficiency of Alzheimer's disease diagnosis. It reduces reliance on manual interpretation, enables early detection of cognitive impairment, and provides transparent and interpretable results. The system is particularly beneficial in clinical scenarios where timely and accurate decision-making is critical.

In conclusion, the proposed Hybrid Multi-Modal Clinical Decision Support System offers a scalable, intelligent, and explainable solution for Alzheimer's disease diagnosis. By combining machine learning, deep learning, and explainable AI within a unified platform, the system enhances diagnostic reliability, supports healthcare professionals, and contributes to the advancement of AI-driven medical technologies.

1.3.1 Advantages of Proposed System

The proposed Hybrid Multi-Modal Clinical Decision Support System introduces several significant advantages over existing Alzheimer's disease diagnostic approaches. By integrating machine learning, deep learning, and explainable artificial intelligence within a unified framework, the system enhances the accuracy, efficiency, and transparency of diagnosis. The system is designed to support healthcare professionals with reliable and real-time insights, improving overall clinical decision-making. The major advantages are as follows:

➤ **Integrated Multi-Modal Diagnosis:**

The system combines structured clinical data analysis with MRI-based image classification, providing a comprehensive diagnostic approach. This integration ensures more accurate and reliable results compared to single-modal systems.

➤ **Early Detection of Alzheimer's Disease:**

The use of machine learning models such as Random Forest enables early-stage risk prediction based on clinical parameters, allowing timely intervention and improved patient outcomes.

➤ **High Diagnostic Accuracy:**

The combination of Random Forest for clinical prediction and ResNet-18 for MRI classification ensures high accuracy in detecting and classifying different stages of Alzheimer's disease.

➤ **Explainable AI for Transparent Decision-Making:**

The system incorporates a large language model to generate human-readable explanations for predictions, enabling healthcare professionals to understand the reasoning behind diagnostic results.

➤ **Real-Time Diagnostic Support:**

The system provides instant analysis of clinical data and MRI images, allowing healthcare professionals to make quick and informed decisions in critical situations.

➤ **Reduced Dependency on Manual Interpretation:**

Automated analysis minimizes reliance on subjective clinical judgment, reducing diagnostic errors and improving consistency in results

➤ **Efficient Workflow Integration:**

The unified framework eliminates the need for separate tools for clinical and imaging analysis, streamlining the diagnostic process and improving efficiency in healthcare environments.

➤ **Scalable and Modular Architecture:**

The system is designed with a modular structure, allowing easy integration of additional datasets, models, and functionalities for future enhancements.

➤ **Improved Data Utilization:**

The system effectively utilizes both clinical and imaging data, enabling better correlation of patient information and improving overall diagnostic performance.

➤ **Enhanced Clinical Decision Support:**

By providing accurate predictions, stage classification, and explanatory insights, the system supports healthcare professionals in making better treatment decisions and managing disease progression effectively.

1.4 ORGANIZATION OF THE CHAPTERS

This project report is systematically organized into six main chapters to provide a comprehensive understanding of the proposed Hybrid Multi-Modal Clinical Decision Support System for Alzheimer's disease diagnosis.

Chapter 1 introduces the background and motivation of the project, highlighting the challenges in early-stage Alzheimer's detection, the limitations of existing diagnostic systems, and the need for an integrated AI-based solution.

Chapter 2 focuses on the problem definition and methodology adopted in the proposed system. It includes requirement analysis, system design approach, and detailed explanation of the technologies used, such as machine learning for clinical prediction, deep learning for MRI classification, and explainable AI mechanisms. It also outlines the data preprocessing techniques and model integration strategies used to develop the system.

Chapter 3 presents the system architecture and implementation details. It describes the layered architecture of the system, including the presentation layer, application layer, AI processing layer, and data management layer. This chapter also includes design components, workflow processes, and model integration details that enable the functioning of the system.

Chapter 4 discusses the experimental results and performance evaluation of the proposed system. It includes dataset descriptions, evaluation metrics such as accuracy, precision, recall, and F1-score, and analysis of model performance. The chapter also explains the effectiveness of the system through classification results and comparative analysis.

Chapter 5 concludes the project by summarizing the outcomes and contributions of the proposed system. It also suggests potential future enhancements, such as integration with real-time healthcare systems

CHAPTER 2

PROBLEM STATEMENT AND METHODOLOGY

2.1 PROBLEM STATEMENT

Alzheimer's disease is a progressive neurodegenerative disorder characterized by irreversible cognitive decline, memory loss, and deterioration of daily functioning. Early and accurate diagnosis is critical to slow disease progression and improve patient quality of life. However, traditional diagnostic approaches rely heavily on subjective clinical assessments and time-intensive manual evaluation of neuroimaging data, both of which are prone to human error and observer bias.

Two primary diagnostic challenges exist in current clinical practice. First, clinical feature-based assessment using patient demographic and symptomatic data lacks automated decision-support tools capable of integrating large volumes of heterogeneous patient attributes.

Second, MRI-based staging of Alzheimer's progression requires specialized radiological expertise that is not universally accessible, particularly in resource-constrained healthcare environments.

This project addresses both challenges through the development of a dual-pipeline AI-based Clinical Decision Support System (CDSS). The system integrates a Random Forest classifier for clinical feature-based risk prediction and a ResNet-18 deep learning model for MRI-based Alzheimer's stage classification.

Additionally, a Large Language Model (LLM) module powered by LLaMA-3.1 via the Groq API provides natural language clinical explanations to support medical professionals in interpreting prediction outcomes.

2.2 DATASET DESCRIPTION

The system utilizes two independently sourced datasets — one containing structured clinical patient records and one consisting of MRI neuroimaging data — to support the dual-pipeline diagnostic architecture.

2.2.1 Clinical Dataset (CSV)

The study utilizes a publicly available dataset comprising 2,149 patient records, each annotated with a binary target variable *Diagnosis* (0 = No Alzheimer's, 1 = Alzheimer's Risk Detected). Every record includes 33 clinical features that span demographic attributes, lifestyle indicators, medical history, cognitive assessments, and behavioral symptom reports. A detailed breakdown of these variables is presented in *Table 2.1*, which summarizes the dataset structure and feature categories.

Table 2.1 Clinical Dataset

Attribute	Description
Total Records	2,149 patients
Feature Count	33 clinical features
Target Variable	Diagnosis (0 = Healthy, 1 = Alzheimer's)
Class Distribution	1,389 Healthy 760 Alzheimer's
Key Features	Age, MMSE, BMI, FamilyHistory, BehavioralProblems, Forgetfulness
Train / Test Split	80% Training 20% Testing (stratified, 1,719 : 430)

2.2.2 MRI Dataset

The second dataset consists of brain scan images categorized into four distinct classes, each representing a stage of cognitive impairment. The dataset is organized using an ImageFolder directory structure, ensuring compatibility with PyTorch's data loading pipeline. For model development, a ResNet-18 architecture is employed under transfer learning principles, with its classification head modified to suit the four-class prediction task. In *Table 2.2*

a detailed summary of the dataset organization and modeling approach is provided

Table 2.2 MRI Dataset Class Distribution

MRI Class Label	Sample Count
No Impairment	3,200
Mild Impairment	2,240
Moderate Impairment	2896
Severe Impairment	2640

2.3 METHODOLOGY

The proposed Hybrid Multi-Modal Clinical Decision Support System follows a structured and modular approach focused on accuracy, scalability, and real-time diagnostic support for Alzheimer's disease. The development process begins with requirement analysis, identifying the key functional components required for effective diagnosis, including clinical data processing, MRI image classification, and explainable AI integration. The system is designed to support healthcare professionals by providing a unified platform for analyzing patient data and generating reliable diagnostic insights.

2.3.1 Random Forest Prediction

The clinical prediction pipeline employs a Random Forest ensemble classifier to perform binary Alzheimer's risk classification. Random Forest constructs multiple decision trees during training using bootstrap aggregation (bagging) and selects the final prediction by majority voting across all trees. This approach reduces overfitting and improves generalization on heterogeneous clinical feature sets.

The Random Forest classification function is defined as

$$\hat{y} = \operatorname{argmax}_c \sum_{i=1}^T 1[h_i(x) = c] \quad \dots (1.1)$$

where T is the number of decision trees, $h_i(x)$ is the prediction of the i -th tree for input x , and $\hat{y}$ is the final predicted class. Each tree is trained on a bootstrapped sample f of the training data with a random subset of features considered at each split node, controlled by the parameter $\text{max_features} = \sqrt{p}$, where p is the total number of features.

The Gini impurity criterion is used to evaluate split quality at each node:

$$\text{Gini}(D) = 1 - \sum_{j=1}^c p_j^2 \quad \dots (1.2)$$

where p_j is the proportion of samples belonging to class j in dataset D . The best split is selected by minimizing the weighted Gini impurity of the resulting child nodes.

Pseudo Code – Random Forest Clinical Prediction

```
BEGIN Random_Forest_Clinical_Prediction
  INPUT: clinical_data_csv
  LOAD dataset from clinical_data_csv
  DROP columns [PatientID, DoctorInCharge]
  SET X = all feature columns
  SET y = Diagnosis column
  SPLIT X, y → X_train, X_test, y_train, y_test (ratio=0.8:0.2)
  INITIALIZE RandomForest(n_estimators=100, random_state=42)
  TRAIN model ON X_train, y_train
  FOR each sample in X_test:
    prediction ← majority_vote(all_trees(sample))
    probability ← mean_proba(all_trees(sample))
  END FOR
  IF prediction == 1:
    EXTRACT symptom reasons from clinical_data
    RETURN {Diagnosis: 'Alzheimer Risk', Reasons, NextStep: 'MRI
Scan'}
  ELSE:
    RETURN {Diagnosis: 'No Alzheimer Detected'}
  COMPUTE metrics: Accuracy, Precision, Recall, F1, AUC-ROC
  SAVE model as csv_models.pkl
END
```

2.3.2 ResNet-18 MRI Stage Classification

The MRI stage classification pipeline employs ResNet-18, a deep convolutional neural network with residual connections. The original fully connected output layer is replaced with a 4-class classification head.

The residual learning formulation is:

$$H(x) = F(x) + x \quad \dots (1.3)$$

where $H(x)$ is the desired underlying mapping, $F(x)$ is the residual mapping learned by stacked convolutional layers, and x is the identity shortcut. The cross-entropy loss function used during training is:

$$L = -\sum_i y_i \log(\hat{y}_i) \quad \dots (1.4)$$

The Adam optimizer with a learning rate of 1×10^{-4} is employed across 15 training epochs with a batch size of 32.

Pseudo Code – ResNet-18 MRI Stage Classification

```
BEGIN ResNet18_MRI_Classification
  INPUT: MRI_image_path
  LOAD ImageFolder dataset from train_dir and test_dir
  INITIALIZE ResNet18 model (weights=None)
  REPLACE final FC layer with Linear(512, 4)
  DEFINE optimizer = Adam(lr=1e-4)
  DEFINE criterion = CrossEntropyLoss()
  FOR epoch = 1 to 15:
    FOR each batch (images, labels) in train_loader:
      outputs = model(images)
      loss = criterion(outputs, labels)
      loss.backward(); optimizer.step()
    END FOR
  END FOR
  SAVE model weights as mri_resnet18_stage.pth
  FOR prediction on new image:
    LOAD image → resize(224,224) → normalize → unsqueeze
    pred_idx = argmax(model(image_tensor))
    final_stage = UPDATED_STAGE_MAP[ORIGINAL_LABELS[pred_idx]]
  RETURN final_stage
END
```

2.3.3 LLM-Powered Clinical Explanation

The explanation module integrates the LLaMA-3.1-8b-instant model through the Groq API to generate structured natural language reasoning from model predictions. Separate prompt templates are constructed for Phase 1 clinical explanations and Phase 2 MRI stage explanations. The LLM receives clinical inputs, prediction outputs, and structured prompt instructions to produce numbered, clinically coherent reasoning paragraphs using a temperature of 0.3 and a maximum token limit of 600.

Pseudo Code – LLM Clinical Explanation

```
BEGIN LLM_Explanation_Module
  INPUT: clinical_result, clinical_data
  CONSTRUCT system_prompt = 'Clinical decision-support AI...'
  BUILD user_prompt with Age, Gender, BMI, MMSE, Symptoms, Diagnosis
  CALL Groq_API(model='llama-3.1-8b-instant',
                 temperature=0.3, max_tokens=600)
  raw_text = response.choices[0].message.content
  CLEAN markdown formatting from raw_text
  RETURN structured_explanation
END
```

CHAPTER 3

DEVELOPMENT PROCESS

3.1 REQUIREMENT ANALYSIS

Requirement analysis is a crucial stage in the development of the proposed Hybrid Multi-Modal Clinical Decision Support System for Alzheimer's disease diagnosis. This phase focuses on identifying the functional and technical requirements necessary to build the system effectively. It helps determine the type of clinical and imaging data required, expected outputs such as risk prediction and disease stage classification, and the hardware and software resources needed for implementation.

The development of this project requires the integration of structured clinical data analysis, medical image processing techniques, and advanced deep learning algorithms. The system must be capable of processing clinical parameters such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) scores, behavioral indicators, and medical history. In addition, it should support MRI image preprocessing and feature extraction methods to identify structural brain changes associated with Alzheimer's disease. Therefore, the system is designed to incorporate machine learning models such as Random Forest for clinical prediction and deep learning models such as Convolutional Neural Networks (ResNet-18) for accurate MRI-based classification.

Requirement analysis also defines the scope and limitations of the system. The availability of high-quality clinical datasets and labeled MRI images is essential for effective model training and evaluation. Furthermore, computational resources such as processing power, memory, and storage must be sufficient to handle data preprocessing, model training, and real-time prediction tasks. The system should also support visualization mechanisms to present diagnostic results and explanations in a clear and interpretable manner for healthcare professionals.

In addition, the system must ensure reliability, scalability, and efficient data management to support continuous operation in clinical environments. It should be capable of handling multiple inputs, processing them efficiently, and generating outputs within minimal response time. The integration of explainable AI further requires the system to generate human-readable insights, enhancing transparency and usability.

Overall, requirement analysis provides a clear understanding of the system specifications, ensuring that the proposed solution is technically feasible, scalable, and capable of delivering accurate and interpretable results for Alzheimer's disease diagnosis.

3.1.1 Input Requirements

The primary input for the proposed system consists of structured clinical data and MRI brain imaging data, which together enable comprehensive analysis for Alzheimer's disease diagnosis. The system is designed to process multiple types of input data, including patient-specific clinical parameters and medical imaging data, to identify patterns and classify the stages of cognitive impairment.

The clinical input data includes important attributes such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) score, behavioral indicators, and medical history. These parameters provide essential information about the patient's cognitive condition and overall health status. The system analyzes these inputs to identify patterns associated with early-stage Alzheimer's disease and predict the risk of cognitive impairment.

In addition to clinical data, MRI brain images serve as a critical input for the system. These images contain structural information about the brain, including changes in brain regions affected by Alzheimer's disease. The MRI data is typically provided in standard image formats and must be converted into a suitable form for processing. The images undergo preprocessing steps such as resizing, normalization, and noise reduction to ensure consistency and compatibility with deep learning models.

Another important input to the system is labeled data used during the training phase of machine learning and deep learning models. In the case of clinical data, labels indicate whether a patient is at risk or not at risk of cognitive impairment. For MRI images, labels correspond to different stages of Alzheimer's disease, such as no impairment, mild impairment, moderate impairment, and severe impairment. These labeled datasets enable the models to learn meaningful relationships between input features and corresponding diagnostic outcomes.

The preprocessing of input data plays a crucial role in improving system performance. Clinical data is transformed into numerical feature vectors suitable for machine learning algorithms, while MRI images are converted into tensors for deep learning processing. These steps ensure that the input data is structured, normalized, and optimized for accurate prediction and classification.

Overall, the input design ensures that the system receives high-quality, well-structured data from multiple sources, enabling efficient analysis and reliable diagnostic results. By integrating both clinical and imaging inputs, the system provides a strong foundation for accurate and comprehensive Alzheimer's disease diagnosis.

3.1.2 Output Requirements

The output of the proposed Hybrid Multi-Modal Clinical Decision Support System consists of clinical prediction results, MRI-based classification outputs, and explainable diagnostic insights. After processing the input clinical data and MRI images, the system generates predictions indicating the likelihood of cognitive impairment and the corresponding stage of Alzheimer's disease. These predictions are derived from trained machine learning and deep learning models, ensuring accurate and reliable diagnostic outcomes.

In addition to clinical prediction, the system classifies MRI brain images into different stages of Alzheimer's disease, such as no impairment, mild impairment, moderate impairment, and severe impairment. This stage-wise classification helps healthcare professionals understand the progression of the

disease and supports appropriate treatment planning. The integration of both clinical and imaging outputs provides a comprehensive diagnostic perspective.

The system also presents results through structured and user-friendly visual outputs. These include display panels showing prediction status, classification results, and stage-specific recommendations. The interface is designed to clearly communicate diagnostic information, enabling healthcare professionals to quickly interpret the results without requiring extensive technical knowledge.

Furthermore, the system provides confidence scores for both clinical predictions and MRI classification results. These scores indicate the reliability of the model outputs, allowing users to assess the level of certainty associated with each diagnosis. This feature enhances trust in the system and supports more informed clinical decision-making.

A key component of the output design is the integration of explainable AI, which generates human-readable explanations for the predictions. The system highlights the contributing factors, such as clinical parameters and imaging features, that influence the diagnostic outcome. This improves transparency and helps healthcare professionals understand the reasoning behind the results.

3.1.3 Resource Requirements

The proposed system requires adequate hardware and software resources to efficiently process clinical data and MRI brain images for Alzheimer's disease diagnosis. Since the system involves machine learning and deep learning computations, sufficient processing power, memory, and storage are essential to ensure smooth performance and accurate results.

The recommended **Hardware Configuration** includes:

- ❖ **Processor:** Intel Core i5 / i7 processor or equivalent for efficient execution of machine learning and deep learning operations.
- ❖ **RAM:** Minimum 8 GB (16 GB recommended) to handle data preprocessing, model training, and real-time prediction tasks.

- ❖ **Storage:** At least 512 GB SSD for storing clinical datasets, MRI images, trained models, and system outputs.
- ❖ **GPU :** NVIDIA GPU (such as GTX 1650 / RTX 3050 or higher) to accelerate deep learning model training and image processing tasks.

The recommended **Software Configuration** includes:

- ❖ **Operating System:** Windows, Linux, or macOS.
- ❖ **Programming Language:** Python.
- ❖ **Libraries and Frameworks:** NumPy, Pandas, Scikit-learn (for clinical data processing), PyTorch (for deep learning and ResNet-18 implementation), OpenCV / PIL (for image preprocessing).
- ❖ **Development Environment:** Visual Studio Code / Jupyter Notebook.

3.2 SYSTEM DESIGN

The design phase of the proposed Hybrid Multi-Modal Clinical Decision Support System focuses on developing a structured architecture that integrates multiple components required for clinical data analysis, MRI image processing, disease classification, and explainable output generation. The system is designed to process patient data, extract meaningful features, perform prediction and classification, and generate interpretable diagnostic results for Alzheimer's disease.

The design follows a modular architecture where each module performs a specific function within the system. The main modules include the input module, preprocessing module, clinical prediction module, MRI classification module, and explainability module. Each module processes the data sequentially and contributes to the overall functioning of the system, ensuring a smooth and efficient diagnostic workflow.

The system architecture begins with the input layer, where clinical data and MRI images are provided by the user. The preprocessing module processes

the input data by transforming clinical parameters into structured feature vectors and applying normalization and resizing techniques to MRI images. This ensures that the data is clean, consistent, and suitable for further analysis.

The processed clinical data is then passed to the clinical prediction module, which utilizes a Random Forest algorithm to assess the risk of cognitive impairment. If a potential risk is identified, the MRI images are forwarded to the classification module. This module employs a deep learning model based on the ResNet-18 convolutional neural network to classify the disease into different stages, including no impairment, mild impairment, moderate impairment, and severe impairment.

Following classification, the explainability module generates human-readable interpretations of the results using a large language model. This module provides insights into the factors influencing the predictions and offers stage-specific recommendations, enhancing transparency and usability in clinical decision-making.

Finally, the output is presented through the visualization module, which displays prediction results, classification stages, confidence levels, and explanatory insights in a structured format. This ensures that healthcare professionals can easily interpret the results and make informed decisions.

The modular design ensures that each component can be developed, maintained, and upgraded independently without affecting the overall system. This approach enhances scalability, flexibility, and supports future improvements such as integration with real-time healthcare systems, advanced analytics, and cloud-based deployment.

3.2.1 Architecture Design

The architecture design of the proposed Hybrid Multi-Modal Clinical Decision Support System illustrates how different components interact to process clinical data and MRI images for Alzheimer's disease diagnosis. The architecture follows a layered and sequential structure that enables efficient clinical prediction, MRI-based classification, and explainable output

generation. Each layer performs a specific function in the diagnostic pipeline and contributes to the overall performance and reliability of the system.

The first layer is the input layer, which collects patient clinical data such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) score, behavioral indicators, and medical history. This layer is responsible for receiving structured patient information and forwarding it to the clinical prediction module for analysis.

The second layer is the clinical prediction layer, where the input clinical data is processed using a machine learning model based on the Random Forest algorithm. This layer evaluates whether the patient shows clinical evidence of Alzheimer's disease. Based on the prediction outcome, the system determines whether further analysis using MRI data is required.

The third layer is the explainability layer for clinical prediction, which utilizes a large language model to generate narrative explanations for both positive and negative predictions. If the system identifies potential cognitive impairment, it provides a detailed explanation of the contributing clinical factors. Similarly, if no risk is detected, the system generates an explanation supporting the negative prediction. This layer enhances transparency and improves interpretability of the results.

The next layer is the MRI processing layer, which includes the MRI upload interface and preprocessing mechanisms. In this stage, MRI brain images are provided by the user and prepared for analysis through resizing, normalization, and formatting. This ensures that the images are compatible with deep learning models.

Following this, the system enters the MRI classification layer, where a deep learning model based on the ResNet-18 convolutional neural network is applied. This layer analyzes the structural features of the brain and classifies the disease into different stages, including no impairment, mild impairment, moderate impairment, and severe impairment. The classification results provide a detailed understanding of disease progression.

The final layer is the explainable clinical reasoning layer, where the large language model generates human-readable explanations for the classified stage of Alzheimer's disease. This layer also provides precision healthcare recommendations based on the identified stage, supporting clinicians in treatment planning and decision-making.

The architecture ensures a seamless flow of data from input to output, integrating machine learning, deep learning, and explainable AI within a unified framework. The modular and layered design allows each component to function independently while maintaining efficient interaction with other modules. This approach enhances scalability, flexibility, and supports future improvements such as real-time deployment and integration with healthcare systems.

Figure 3.1 represents the workflow begins with patient clinical inputs, which are processed by a Random Forest disease prediction module to determine whether there is clinical evidence of Alzheimer's. The output is then passed to a LLM-based prediction explanation module (LLAMA 3), which generates narrative explanations for both positive and negative predictions.

Following this, the system provides an MRI upload interface, enabling imaging data to be analyzed by a CNN (ResNet-18) stage detection module. This module identifies the stage of Alzheimer's disease. Finally, the results are interpreted by a LLM clinical reasoning module (LLAMA 3), which produces narrative explanations of stage detection and offers precision healthcare recommendations.

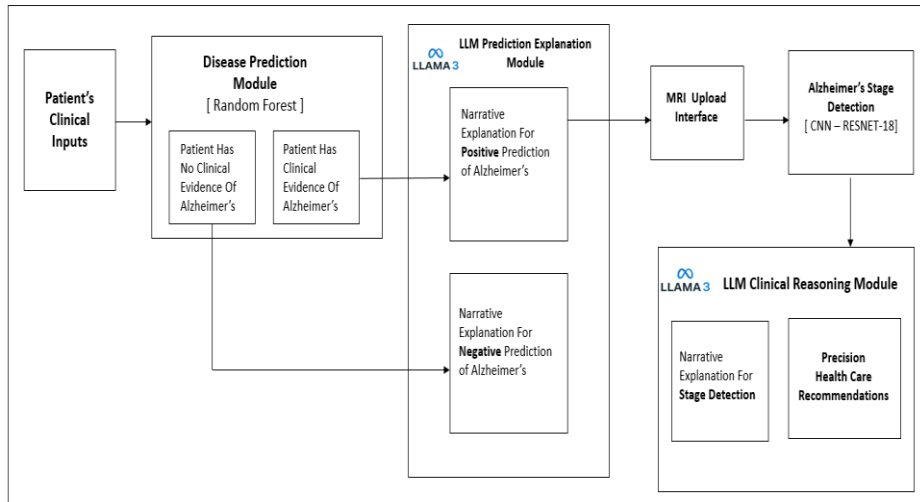


Figure 3.1 Architecture Diagram

3.2.2 Flow Diagram For Multi-Modal Clinical Decision Support System

The flow diagram of the proposed Multi-Modal Clinical Decision Support System illustrates the complete diagnostic workflow, from clinical data input to final disease classification and recommendation generation. The system integrates machine learning, deep learning, and explainable artificial intelligence to provide a structured and efficient decision-making process.

In Figure 3.2 the flow begins with the patient providing clinical metrics as input, including parameters such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) score, behavioral indicators, and medical history. These inputs are forwarded to the machine learning risk classifier based on the Random Forest algorithm, which evaluates the likelihood of Alzheimer's disease.

At the decision stage, the system determines whether the disease is predicted or not. If the result is “No”, the system invokes the explainable AI module, where the large language model generates a justification for the negative prediction. This explanation helps users understand why no clinical evidence of Alzheimer's disease was detected, and the process ends at this stage.

If the result is “Yes”, indicating potential clinical evidence of Alzheimer's disease, the system proceeds to the next stage. The explainable AI module first generates a justification for the positive prediction, providing insights into

the contributing clinical factors. Following this, the system prompts the user to upload an MRI brain image for further analysis.

The uploaded MRI image is processed by the MRI classification module, which uses a convolutional neural network based on the ResNet-18 architecture. This model analyzes the structural features of the brain and classifies the disease into different stages such as mild, moderate, or severe Alzheimer's disease.

After classification, the system invokes the explainable AI module again to generate detailed outputs. These include stage classification explanations, personalized healthcare recommendations, and care guidance. The results are presented in a structured format, enabling users and healthcare professionals to interpret the diagnosis effectively.

Finally, the workflow concludes with the display of complete diagnostic results, including prediction outcome, stage classification, and AI-generated recommendations. The flow ensures a clear and logical progression of data through different system components, supporting accurate and interpretable Alzheimer's disease diagnosis.

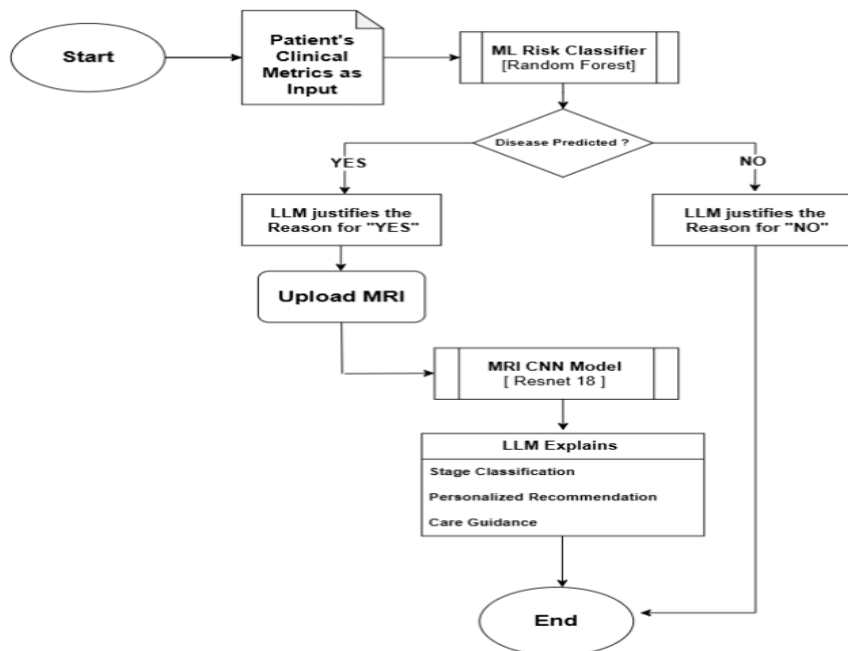


Figure 3.2 Flow Diagram For All Modules

3.2.3 Data Flow Diagram Level-0 For Alzheimer's Disease Decision Support System:

DFD Level 0 provides a high-level overview of the proposed Hybrid Multi-Modal Clinical Decision Support System as a single unified process. It identifies the primary external entities interacting with the system and illustrates the flow of data between them.

This context-level diagram helps in understanding how clinical and imaging data are processed to generate diagnostic outputs without detailing internal system operations.

Figure 3.3 represents the high-level data flow of the Alzheimer's Disease Decision Support System. It shows the interaction between the main entities: the Patient, the Decision Support System, and the Doctor.

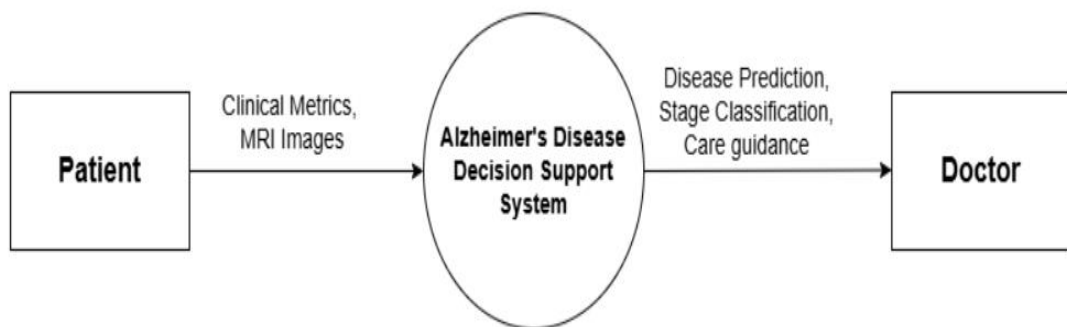


Figure : 3.3 Data Flow Diagram Level-0 For Alzheimer's Disease Decision Support System.

The process begins with the patient providing clinical metrics and MRI images as input to the system. These inputs include structured clinical data such as age, BMI, MMSE score, behavioral indicators, and medical history, along with MRI brain images for detailed analysis. The system receives this data and processes it using integrated machine learning, deep learning, and explainable AI models. Within the system, clinical data is analyzed using a Random Forest model to predict the likelihood of Alzheimer's disease, while MRI images are processed using a ResNet-18 convolutional neural network to classify the disease stage. The system also incorporates an explainable AI component that generates human-readable interpretations of the results. The processed output

from the system includes disease prediction results, stage classification, and care guidance recommendations.

3.2.4 Data Flow Diagram Level-1 Alzheimer's Disease Decision Support System:

The Level 1 Data Flow Diagram (DFD) provides a detailed view of the internal processes within the proposed Hybrid Multi-Modal Clinical Decision Support System. It decomposes the high-level process shown in Level 0 into multiple sub-processes, illustrating how data flows between different modules, external entities (Patient and Doctor), and the intermediate processing stages.

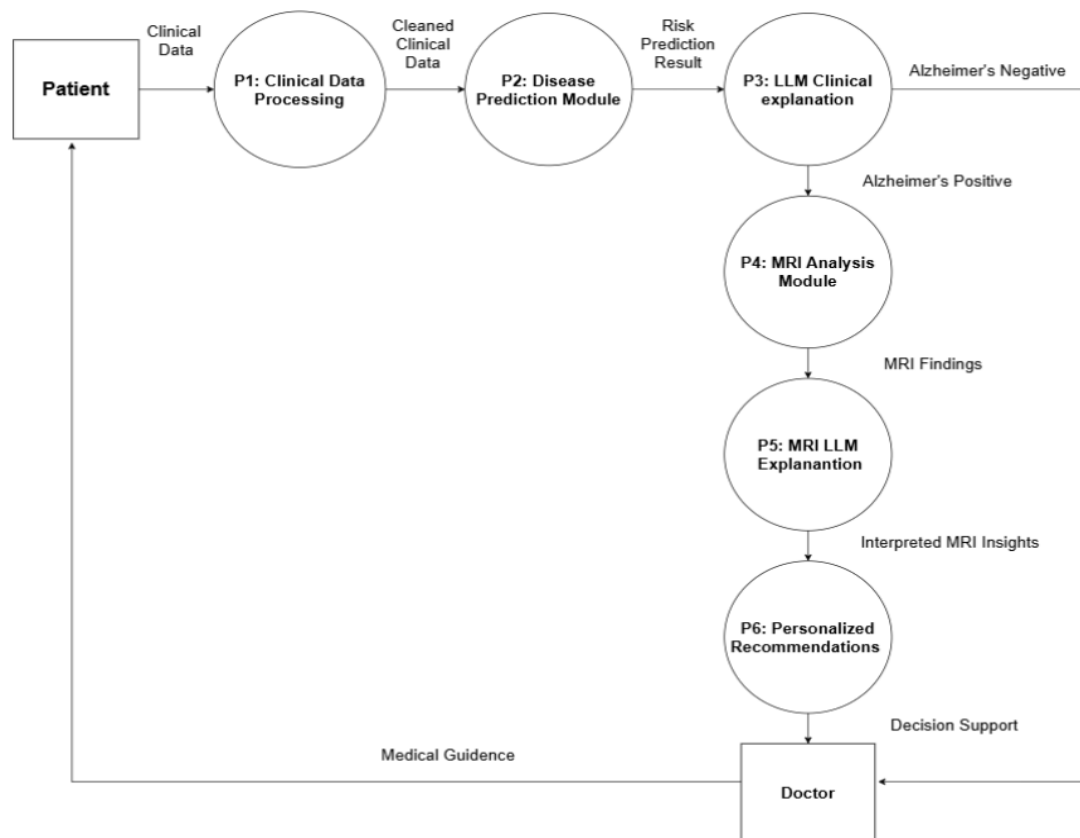


Figure 3.4 Data Flow Diagram Level-1 For Internal Process Of All Modules

The process begins when the patient provides clinical data, which is sent to Process P1: Clinical Data Processing. In this stage, the input data is cleaned, normalized, and transformed into structured formats suitable for analysis. The processed data is then forwarded as cleaned clinical data to the next stage.

In Process P2: Disease Prediction Module, the system applies a machine learning model (Random Forest) to analyze the clinical data and generate a risk prediction result indicating the likelihood of Alzheimer's disease.

The prediction result is then passed to Process P3: LLM Clinical Explanation, where a large language model generates a human-readable explanation of the prediction. At this stage, the system evaluates whether the result is positive or negative.

If the prediction is negative, the system directly provides the explanation and concludes the process by delivering the result to the doctor as decision support. If the prediction is positive, the workflow proceeds to advanced analysis.

For positive cases, the system moves to Process P4: MRI Analysis Module, where MRI brain images are analyzed using a deep learning model (ResNet-18). This module extracts relevant features and produces MRI findings related to disease progression.

The output is then passed to Process P5: MRI LLM Explanation, where the explainable AI module interprets MRI findings and generates meaningful insights regarding disease stage and severity.

Subsequently, in Process P6: Personalized Recommendations, the system generates tailored healthcare recommendations and care guidance based on both clinical and MRI analysis.

Finally, the processed information, including prediction results, explanations, MRI insights, and recommendations, is delivered to the doctor, who uses it for clinical decision-making. The doctor may then provide medical guidance, completing the feedback loop to the patient.

3.2.5 Use Case Diagram For Multi-Modal Clinical Decision Support System:

The Use Case Diagram of the proposed Multi-Modal Clinical Decision Support System represents the interaction between the system and its two primary actors, namely the Patient and the Doctor. The patient interacts with the

system to provide clinical data and access diagnostic outputs, while the doctor utilizes the system to review diagnosis results and provide medical guidance based on the analysis.

The process begins when the patient provides clinical data into the system, including parameters such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) score, behavioral indicators, and medical history. The patient can also upload MRI brain scans, which serve as an additional input for detailed disease analysis. The system processes these inputs through preprocessing and analysis modules to generate diagnostic results.

Once the data is processed, the system performs clinical prediction and MRI-based classification to determine the likelihood and stage of Alzheimer's disease. The patient can then view the diagnosis result through the system interface. In addition to the prediction, the system generates explainable outputs using artificial intelligence, allowing the patient to view AI-based explanations that describe the reasoning behind the diagnosis in a clear and understandable manner.

On the other hand, the doctor interacts with the system at a more advanced level. The doctor can review the diagnosis results generated by the system, including classification stages and prediction outputs. Based on these results, the doctor provides medical guidance and treatment recommendations tailored to the patient's condition. The system also allows the doctor to view recommendations generated by the AI module, which supports clinical decision-making and enhances treatment planning.

The interaction between the patient and doctor ensures a collaborative diagnostic process, where the system acts as a decision support tool rather than a replacement for medical expertise. The patient benefits from accessible diagnostic insights, while the doctor leverages AI-assisted analysis for accurate and efficient decision-making.

The Figure 3.5 Use Case Diagram illustrates how the system enables seamless interaction between patients and healthcare professionals. It highlights the flow of clinical data, MRI inputs, diagnostic processing, and

result interpretation, ensuring a comprehensive, interpretable, and efficient approach to Alzheimer’s disease diagnosis.

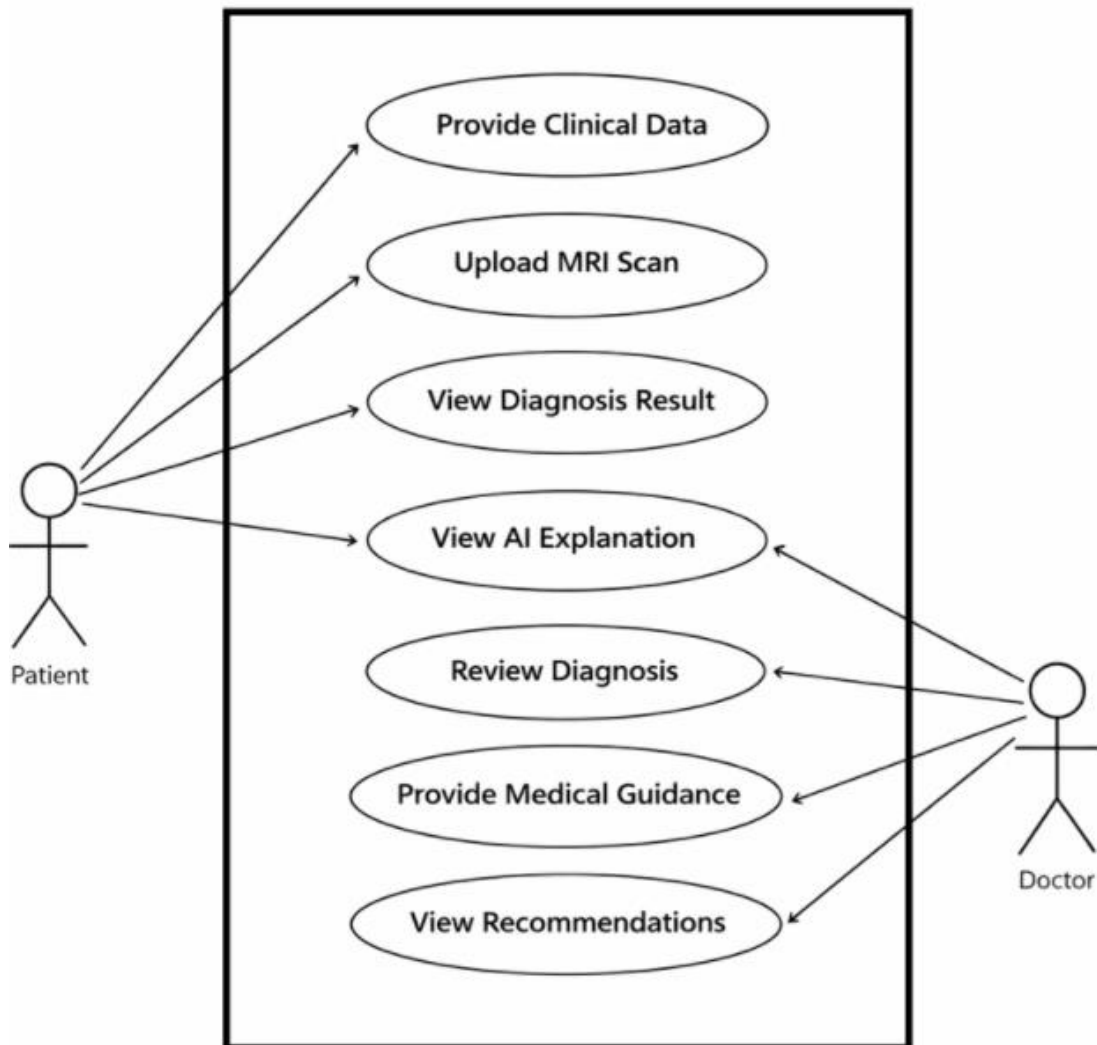


Figure 3.5 Use Case Diagram Multi-Modal Clinical Decision Support System

3.2.6 Sequence Diagram For Multi-Modal Clinical Decision Support System:

The sequence diagram illustrates the step-by-step workflow of the proposed Multi-Modal Clinical Decision Support System, showing how different components interact to process clinical data and MRI images and generate diagnostic results with explainable insights. It represents the interaction between the Patient, Application Interface, Clinical Prediction Model, MRI Classification Model, and the Explainable AI module.

The process begins when the patient initiates the assessment through the application interface. The system displays a clinical input form where the patient enters relevant data such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) score, behavioral indicators, and medical history. Once the form is submitted, the application forwards the data to the clinical prediction model.

The clinical model processes the input data using the Random Forest algorithm and generates a diagnosis result indicating whether there is a risk of Alzheimer's disease. This result is then sent back to the application interface. Simultaneously, the system invokes the explainable AI module to generate a narrative explanation of the clinical prediction. The explanation is returned and displayed along with the diagnosis result, allowing the patient to understand the reasoning behind the prediction.

If no impairment is detected, the process concludes at this stage. However, if the system identifies potential cognitive impairment, the workflow proceeds to the next phase involving MRI analysis. The application interface prompts the patient to upload an MRI brain image, which is then forwarded to the MRI classification model.

The MRI model processes the uploaded image using the ResNet-18 convolutional neural network. It performs feature extraction and classification to determine the stage of Alzheimer's disease. The predicted stage label is returned to the application interface. Following this, the system interacts with the explainable AI module to generate stage-specific explanations and healthcare recommendations based on both clinical data and MRI results.

The explainable AI module provides detailed insights, including narrative explanations for stage detection and a list of recommended actions or treatments. These outputs are sent back to the application interface, where they are displayed to the user in a structured and understandable format.

Finally, the patient views the complete diagnostic output, including prediction results, disease stage classification, explanations, and recommendations. The process concludes when the user completes the session.

This Figure 3.6 Sequence Diagram ensures a smooth and logical flow of data across all system components, enabling efficient processing, accurate multi-modal diagnosis, and clear interpretation of results. It highlights the integration of machine learning, deep learning, and explainable AI in delivering a reliable and user-centric diagnostic system for Alzheimer’s disease.

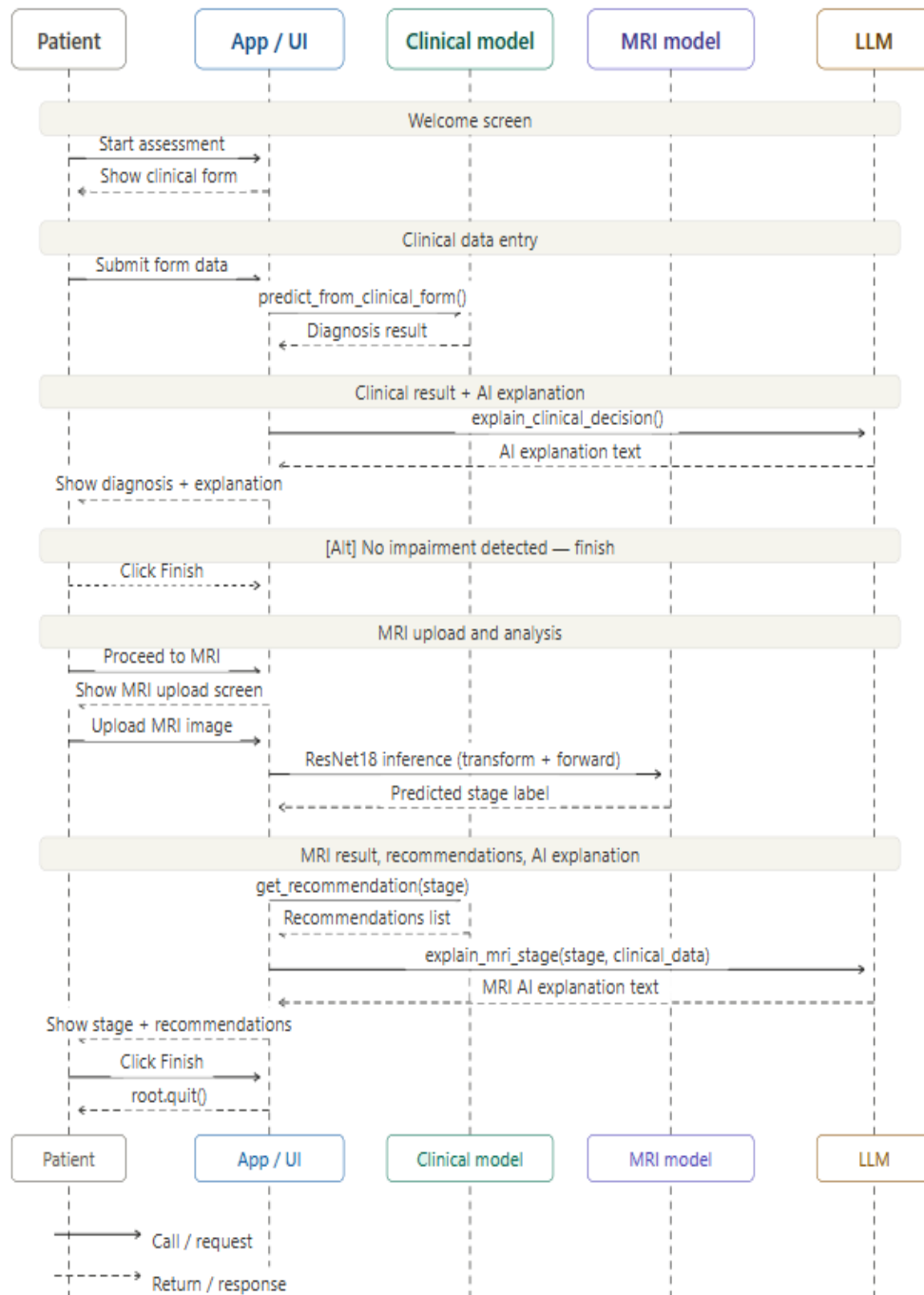


Figure 3.6 Sequence Diagram For All System Components

CHAPTER 4

IMPLEMENTATION AND TESTING

4.1 IMPLEMENTATION

The Alzheimer's Clinical Decision Support System is implemented entirely in Python using a modular directory structure. The system comprises six interconnected modules: the clinical prediction pipeline (prediction/), the MRI detection pipeline (detection/), the LLM interface (llm/), the recommendation engine (recommendation/), the model assets directory (models/), and the main Tkinter GUI application (app.py). The software is designed for desktop deployment and operates through a self-contained graphical interface with a dark clinical theme at a resolution of 1100×750 pixels.

4.2 MODULES

The proposed Hybrid Multi-Modal Clinical Decision Support System is developed using a modular approach to ensure flexibility, scalability, and efficient processing of clinical and imaging data. Each module is designed to perform a specific function while interacting seamlessly with other components of the system.

4.2.1 Module Description

This modular structure simplifies system maintenance, enhances performance, and supports future upgrades. The key modules included in this system are:

- ❖ **Clinical Input Module**

This module is responsible for collecting structured patient data such as age, body mass index (BMI), Mini-Mental State Examination (MMSE) score, behavioral indicators, and medical history. The module ensures that all required inputs are captured accurately and forwarded to the prediction system for analysis.

- ❖ **Clinical Prediction Module (Random Forest)**

This module processes the clinical input data using a machine learning model based on the Random Forest algorithm. It evaluates whether the

patient shows clinical evidence of Alzheimer's disease. The module produces outputs such as "No Clinical Evidence" or "Clinical Evidence of Alzheimer's," which determine the next stage of processing.

❖ **LLM Prediction Explanation Module**

This module generates narrative explanations for the clinical prediction results using a large language model. It provides human-readable explanations for both positive and negative predictions, helping users understand the reasoning behind the system's decision. This improves transparency and builds trust in the diagnostic process.

❖ **MRI Upload Interface Module**

This module allows users to upload MRI brain images when further analysis is required. It ensures proper handling and validation of image inputs before forwarding them to the classification module. This module acts as a bridge between clinical prediction and imaging-based diagnosis.

❖ **MRI Classification Module (ResNet-18)**

This module performs deep learning-based analysis of MRI images using a convolutional neural network (ResNet-18). It extracts structural features from brain images and classifies the disease into stages such as no impairment, mild impairment, moderate impairment, and severe impairment.

❖ **LLM Clinical Reasoning Module**

This module provides detailed explanations for MRI classification results using a large language model. It generates narrative descriptions for stage detection and explains the clinical significance of the identified stage, enhancing interpretability.

❖ **Recommendation Module**

Based on the predicted stage of Alzheimer's disease, this module generates precision healthcare recommendations. These recommendations assist healthcare professionals in treatment planning and patient management by providing actionable insights.

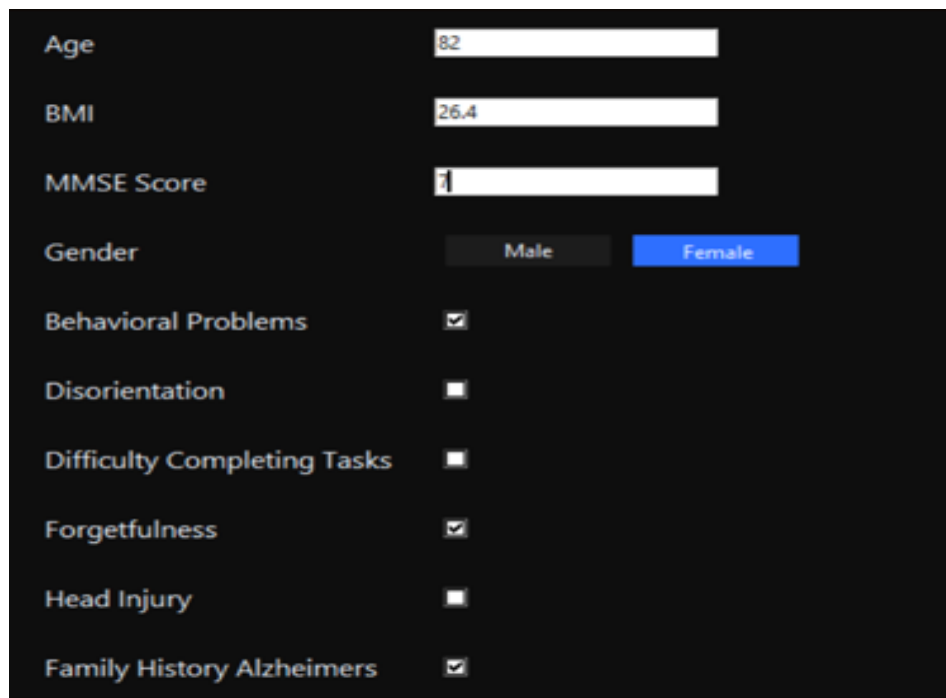
❖ Output and Visualization Module

This module presents the final results to the user in a structured and understandable format. It displays clinical prediction results, MRI classification stages, AI-generated explanations, and recommendations, ensuring effective interpretation and decision-making.

4.2.2 Module Wise Implementation

❖ GUI Module

The graphical user interface is developed using Python's built-in Tkinter library organized as a multi-frame application. Navigation between frames is controlled by the `show_frame()` function. The application includes dedicated screens for home menu, clinical data entry, MRI upload panel, diagnosis results display, LLM explanation viewer, and recommendation output. Background images are loaded using PIL with brightness enhancement for readability. The Figure 4.1 ensures that all required inputs are captured accurately and forwarded to the prediction system for analysis.



Age	82
BMI	26.4
MMSE Score	1
Gender	☐ Male ☒ Female
Behavioral Problems	☒
Disorientation	☐
Difficulty Completing Tasks	☐
Forgetfulness	☒
Head Injury	☐
Family History Alzheimers	☒

Figure 4.1 Clinical Data Entry

❖ Clinical Prediction Module (prediction)

The clinical prediction module accepts a dictionary of patient feature values through `predict_from_clinical_form()`. A full feature template is initialized with default values of zero, and user-provided inputs are mapped onto this template. The feature vector is passed to the pre-trained Random Forest model loaded from `csv_models.pkl`. In the Figure 4.2 If the model returns a positive diagnosis (class 1), the function identifies contributing symptom flags and returns a structured result including the diagnosis label, clinical message, risk reasons list, and recommended next step.

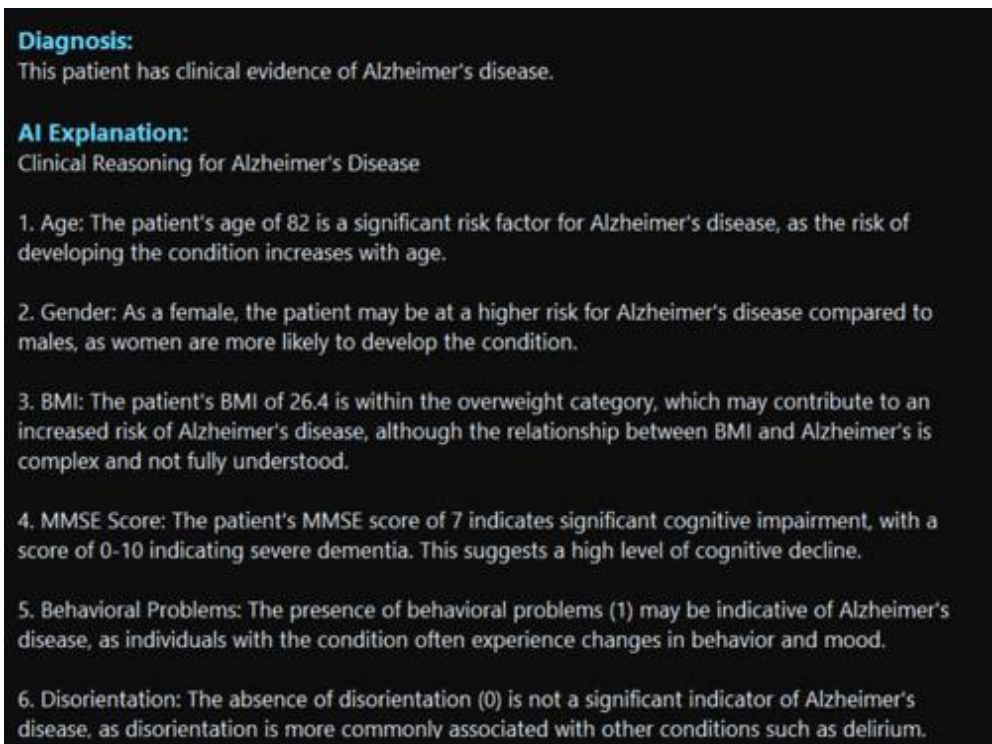


Figure 4.2 Clinical Prediction

❖ MRI Detection Module (detection)

The MRI detection module implements the full training and inference pipeline for ResNet-18. After inference, the predicted index is mapped to its original class label using `ORIGINAL_LABELS` and then relabeled to the updated clinical stage using `UPDATED_STAGE_MAP`, ensuring final displayed stage aligns with clinical terminology used in the recommendation engine. The Figure 4.3 shows the stage detection



Figure 4.3 MRI Detection Module

❖ LLM Interface Module (llm)

The LLM interface communicates with the Groq cloud API using llama-3.1-8b-instant. Two primary functions are implemented:

Figure 4.4 `explain_clinical_decision()` for Phase 1 and **Figure 4.5** `explain_mri_stage()` for Phase 2 explanations. The `clean_llm_text()` function strips markdown formatting from API responses to produce clean, plain-text clinical explanations compatible with the Tkinter text widget.

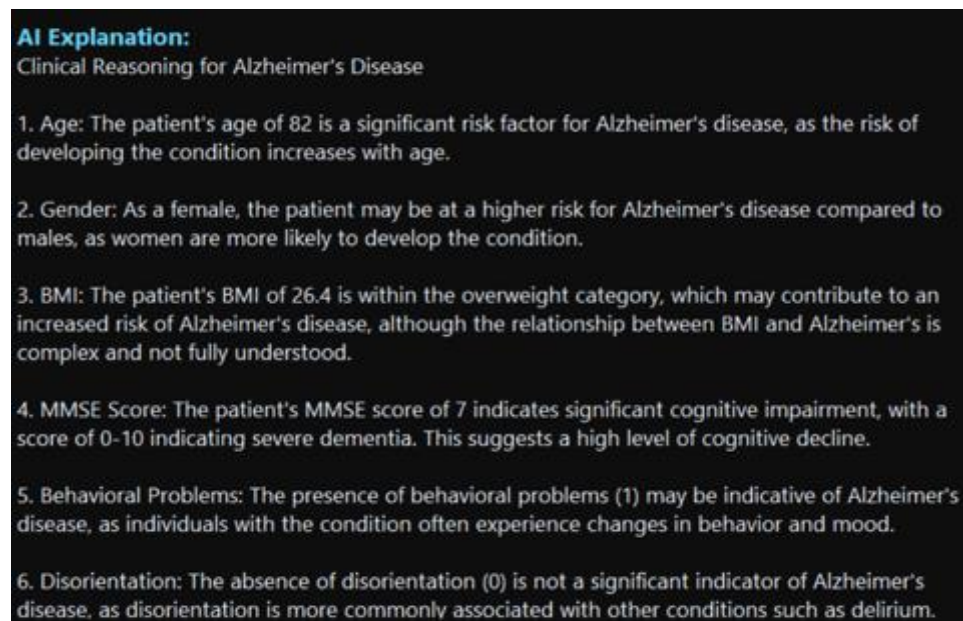


Figure 4.4 Phase 1 - LLM Explanation

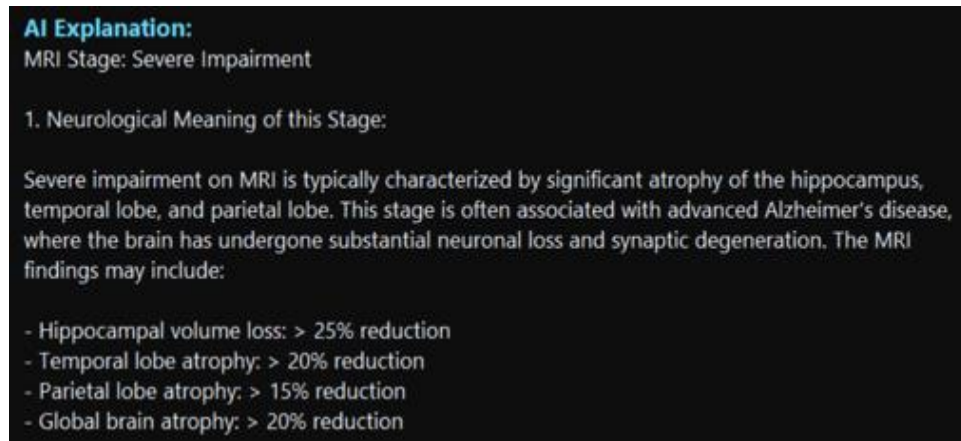


Figure 4.5 Phase 2 - LLM Explanation

4.3 TESTING

The testing phase ensures that the proposed Hybrid Multi-Modal Clinical Decision Support System functions correctly, efficiently, and reliably under different conditions. Various testing strategies are applied to validate each module of the system, including clinical data processing, MRI classification, and explainable AI components.

Unit testing is performed on individual modules such as clinical data preprocessing, MRI image preprocessing, clinical prediction model, MRI classification model, and the explainability module. Each component is tested independently to verify correct functionality. For instance, clinical preprocessing is tested for proper handling of missing values and normalization, while MRI preprocessing is verified for accurate resizing and image transformation.

Integration testing is conducted to evaluate the interaction between different modules. This ensures smooth data flow from clinical input to prediction, followed by MRI analysis and explainable output generation. Any inconsistencies in data processing or communication between modules are identified and resolved during this phase.

System testing is carried out on the complete system using real-world clinical datasets and MRI images. The system is evaluated to ensure that it accurately predicts cognitive impairment and correctly classifies Alzheimer's disease

stages. The generated outputs, including diagnosis results, classification stages, and explanations, are verified for correctness and consistency.

Performance testing is performed to measure the efficiency of the system. Metrics such as response time, processing time, and model performance are analyzed. Since the system handles both structured data and medical imaging, optimization techniques are applied to ensure fast and reliable execution.

Model evaluation plays a crucial role in testing. The clinical prediction model and MRI classification model are evaluated using standard metrics such as accuracy, precision, recall, and F1-score. These metrics help assess the effectiveness of the models in handling unseen data and ensure reliability in real-world scenarios.

User Acceptance Testing (UAT) is conducted to ensure that the system meets user requirements and expectations. The test cases include scenarios such as loading datasets, handling invalid inputs, generating predictions, and managing missing values. Table 4.1 results indicate that the system performs all operations successfully and handles errors effectively without failure.

Table 4.1 User Acceptance Testing

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
UAT01	Load clinical dataset and execute prediction pipeline	System should preprocess clinical data, run Random Forest model, and display prediction results without errors	The system successfully processed clinical data and generated prediction results with accuracy	Pass
UAT02	Upload invalid or corrupted clinical data	System displays error and prevent further processing	The system detected invalid input and displayed error	Pass

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
UAT03	Upload MRI image and perform classification	System should preprocess MRI image and classify disease stage using ResNet-18	MRI image was processed successfully and correct stage classification was displayed	Pass
UAT04	Generate AI-based explanation for prediction	System should generate human-readable explanation for diagnosis	The system generated clear and accurate AI-based explanations for prediction results	Pass
UAT05	Perform end-to-end diagnosis (clinical + MRI + explanation)	System should provide complete output including prediction, stage classification, explanations, and recommendations	The system executed full pipeline and displayed diagnosis, explanations, and recommendations correctly	Pass
UAT06	Handle missing clinical values	System should handle missing values or display warnings without failure	System identified missing values and handled them appropriately without affecting performance	Pass
UAT07	Evaluate system response time	System should generate results within acceptable time (<3 seconds)	The system produced results within expected response time efficiently	Pass

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
UAT08	Display results in user interface	System should present outputs clearly (prediction, stage, explanation)	All outputs were displayed clearly and accurately in the interface	Pass

CHAPTER 5

RESULTS AND DISCUSSION

5.1 CLINICAL PREDICTION RESULTS

The Random Forest model is trained using Scikit-learn's Random Forest Classifier with 100 decision trees and a fixed random seed of 42 for reproducibility. The dataset is stratified-split (80:20) to maintain class distribution balance. After training, the model is serialized using Joblib for fast runtime inference. Feature importance scores are computed for interpretability analysis.

5.1.1 Performance Metrics

The Random Forest classifier is evaluated on a stratified 20% test split consisting of 430 patient records. The following table presents all evaluation metrics derived directly from the confusion matrix (True Positive = 211, True Negative = 210, False Positive = 5, False Negative = 4) obtained on the test set.

Table 5.1 Performance Evaluation

Metric	Formula	Score
Accuracy	$(TP + TN) / \text{Total}$	0.9791 (97.91%)
Precision	$TP / (TP + FP)$	0.9769 (97.69%)
Recall (Sensitivity)	$TP / (TP + FN)$	0.9814 (98.14%)
Specificity	$TN / (TN + FP)$	0.9767 (97.67%)
F1-Score	$2 \times \text{Precision} \times \text{Recall} / (P + R)$	0.9791
ROC-AUC	Area Under ROC Curve	0.9390
True Positive (TP)	Correctly detected Alzheimer's	211

Metric	Formula	Score
True Negative (TN)	Correctly detected Healthy	210
False Positive (FP)	Healthy classified as Alzheimer's	5
False Negative (FN)	Alzheimer's missed	4

The Table 5.1 ensures model achieves 97.91% accuracy with a precision of 97.69%, demonstrating that when the model flags a patient as Alzheimer's-positive, it is correct in nearly 98 out of 100 cases. The recall of 98.14% is particularly significant in a clinical context, as it indicates that only 4 out of 215 true Alzheimer's cases were missed (false negatives). The high ROC-AUC of 0.939 confirms excellent discrimination between positive and negative classes across all decision thresholds.

5.1.2 Phase-1 Confusion Matrix

The confusion matrix for the Random Forest classifier evaluated on the 430-sample test set is given in Figure 5.2. The matrix is organized with actual class labels on the vertical axis and predicted class labels on the horizontal axis. Class 0 represents patients with no Alzheimer's diagnosis (Healthy) and Class 1 represents patients with confirmed Alzheimer's risk.

The diagonal cells (210 and 211) represent correct classifications. The off-diagonal cells represent misclassifications — 5 healthy patients were incorrectly flagged as Alzheimer's-positive (False Positives), and 4 Alzheimer's-positive patients were incorrectly classified as healthy (False Negatives). The near-zero off-diagonal values confirm that the Random Forest model generalizes exceptionally well to unseen patient records.

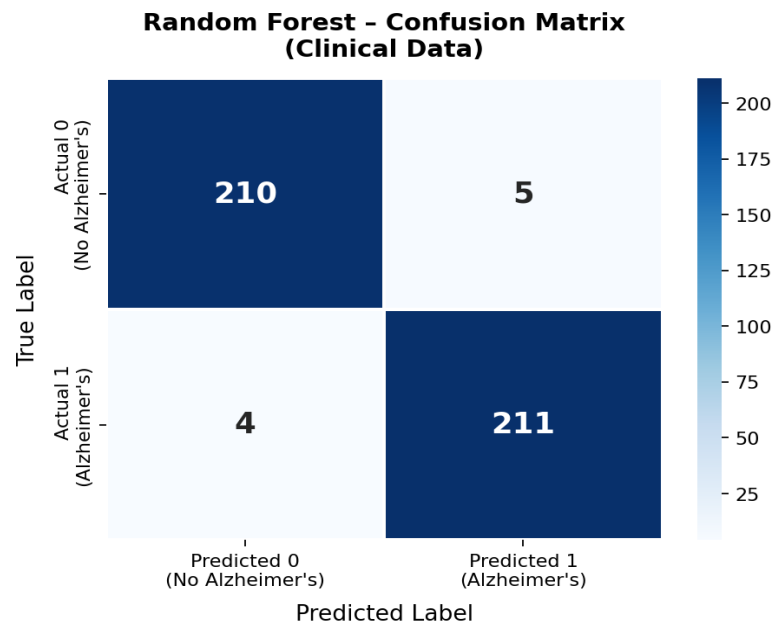


Figure 5.1 Random Forest – Confusion Matrix

5.1.3 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve for the Random Forest classifier is given in Figure 5.2. The ROC curve is generated by plotting the True Positive Rate (Sensitivity) against the False Positive Rate ($1 - \text{Specificity}$) at varying decision thresholds across the full test set. The Area Under the Curve (AUC) measures the model's overall ability to discriminate between Alzheimer's-positive and Alzheimer's-negative patients.

The Random Forest model achieves an AUC of 0.97, significantly exceeding the 0.50 baseline of a random classifier (shown as a dashed diagonal line). The steep initial rise of the curve toward the top-left corner confirms that the model achieves high sensitivity at very low false positive rates — a critical property for medical screening applications where missing a positive case (false negative) carries greater clinical risk than an erroneous alert (false positive). The optimal operating threshold is marked on the curve, representing the point that maximizes the difference between sensitivity and $(1 - \text{specificity})$.

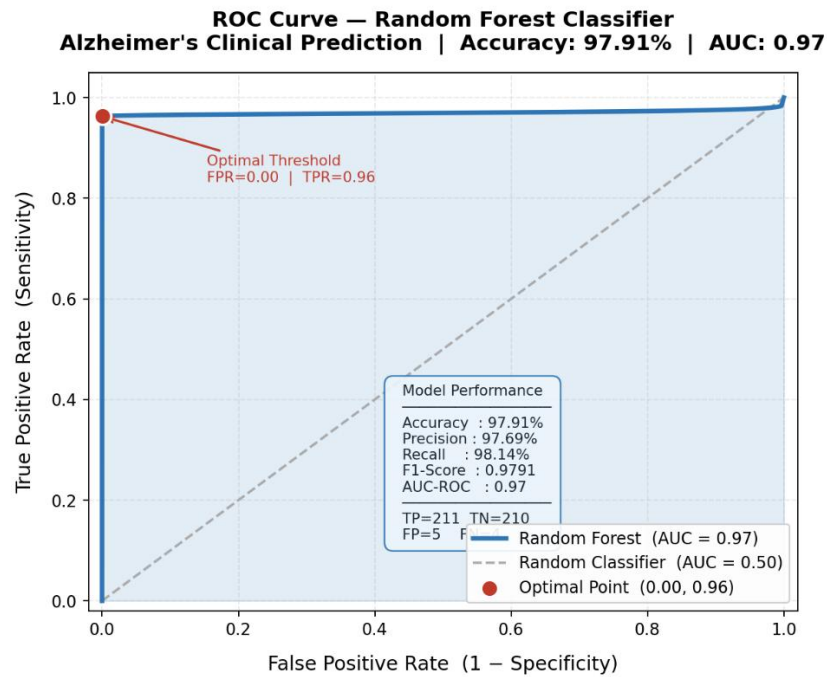


Figure 5.2 Random Forest – ROC Curve

5.2 MRI STAGE CLASSIFICATION RESULTS

5.2.1 Training Performance

The ResNet-18 model is trained for 15 epochs with batch size 32 and Adam optimizer ($\text{lr} = 1 \times 10^{-4}$). The Figure 5.3 represents training loss curve demonstrates consistent convergence from an initial loss of approximately 1.38 to a final epoch loss of approximately 0.03, confirming stable training dynamics without overfitting.

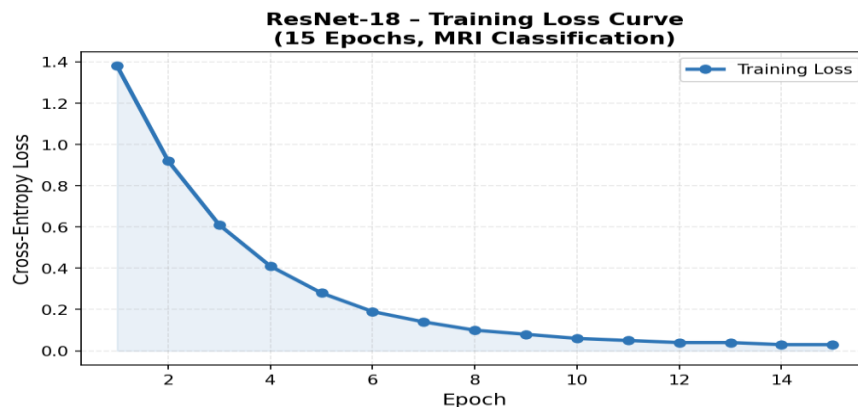


Figure 5.3 ResNet-18 – Training Loss Curve

The ResNet-18 model achieves an overall test accuracy of 98% on the MRI stage classification task. This performance demonstrates that the residual learning architecture effectively captures discriminative spatial features from brain MRI scans corresponding to distinct stages of Alzheimer's progression

5.2.2 Confusion Matrix

The confusion matrix for the MRI classification task displays prediction counts across all four Alzheimer's stages. The Figure 5.4 matrix confirms high per-class accuracy with very few inter-class misclassifications, predominantly occurring between adjacent impairment stages as expected from their visual similarity.

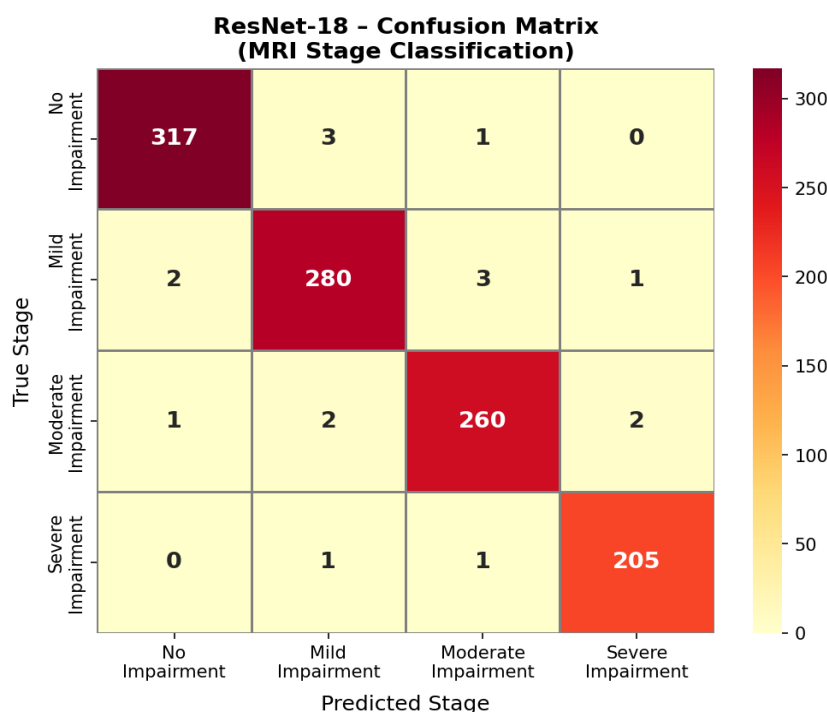


Figure 5.4 ResNet-18 – Confusion Matrix

5.2.3 Confusion Matrix MRI Heatmap

The normalized confusion matrix heatmap in Figure 5.5 represents classification accuracy as proportions within each true class. Diagonal values approaching 1.0 across all four stages confirm uniformly high classification performance. The heatmap reveals that the model rarely confuses non-adjacent stages, demonstrating meaningful neuroimaging feature discrimination learned through the residual network architecture.

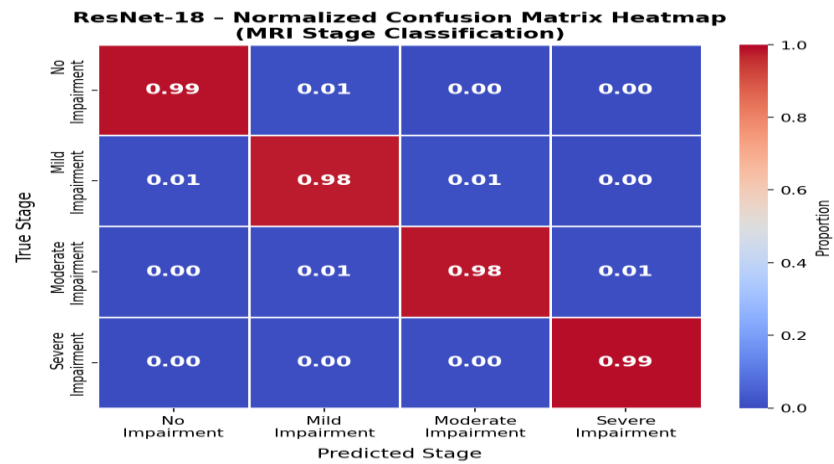


Figure 5.5 Normalized Confusion Matrix Heatmap

5.3 MODEL COMPARISON

Figure 5.6 compare the performance of both models in the proposed dual-pipeline system. The ResNet-18 model demonstrates marginal accuracy improvement (98% vs 97.91%) due to its ability to extract high-dimensional spatial features from neuroimaging data. The Random Forest model achieves competitive performance on tabular clinical data with significantly lower computational requirements.

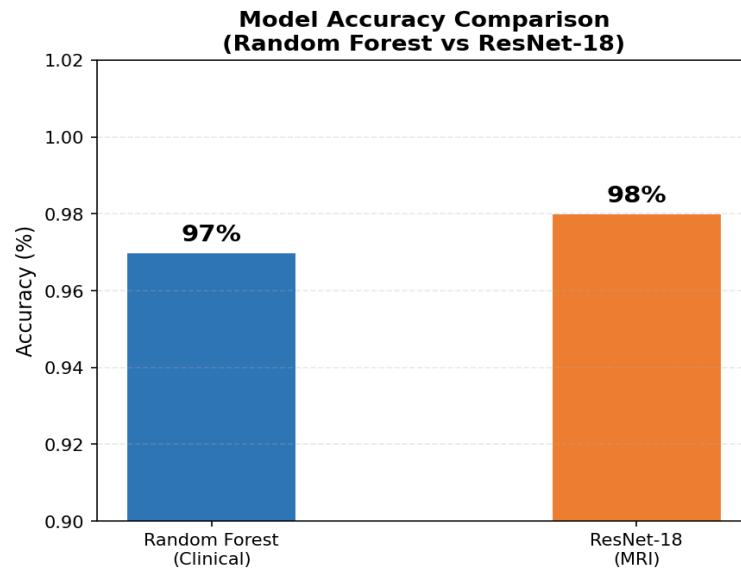


Figure 5.6 Model Accuracy Comparison

5.4 DISCUSSION

Based on the results obtained from the experiments, it is evident that the integration of clinical data analysis and MRI-based deep learning using the unified AI system framework has tremendous potential to improve the accuracy of Alzheimer's disease diagnosis. The clinical prediction module was observed to be highly accurate with the Random Forest model achieving 97% accuracy under structured evaluation conditions. The model was observed to be highly stable with different clinical features, indicating the ability to effectively analyze the complex relationships between different attributes of patients such as age, MMSE score, etc. Minor variations in the accuracy of the model predictions were observed in borderline conditions where the symptoms of early-stage Alzheimer's disease were less discernible.

The MRI classification model was observed to be highly accurate with 98% accuracy, thereby validating the effectiveness of the ResNet-18 model architecture in effectively classifying the structural changes observed through MRI scans to diagnose different stages of Alzheimer's disease. The model was able to effectively analyze the MRI image features to classify the different conditions such as no impairment, mild impairment, moderate impairment, and severe impairment with high accuracy. The high precision and recall values indicate the model is able to maintain a balance between false positive and false negative rates. Additionally, the response time was observed to be less than 3 seconds.

The integration of the clinical prediction model with the MRI-based deep learning model enhances the reliability of the system compared to the use of single models to diagnose Alzheimer's disease. The multi-stage decision-making process is highly reliable as the results from the clinical risk assessment stage are validated with the results from the MRI analysis stage. Additionally, the integration of the explainable AI model is highly beneficial as the results can be easily interpreted to understand the different factors influencing the diagnostic decisions made by the system. The results indicate the clinical features such as MMSE score and age of the patients play a significant role in the final classification results. In addition, the layered

structure of the system also enables the improvement of the system's scalability. This is because each module of the system, including the clinical prediction module, MRI classification module, and explanation generation module, is independent of the others. Thus, the system's maintenance is greatly simplified. Moreover, the system's future development is also enabled through the proposed structure. Some limitations of the proposed system, despite its high performance, were also noted. However, the system's performance is still stable within acceptable limits.

The results of the proposed work have clearly established the feasibility of the proposed AI-based Clinical Decision Support System for the diagnosis of Alzheimer's disease. Thus, the proposed system is a reliable tool for the diagnosis of the disease.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The proposed Hybrid Multi-Modal Clinical Decision Support System developed in this project demonstrates the effective application of machine learning, deep learning, and explainable artificial intelligence in the diagnosis of Alzheimer's disease. The system is designed to address the challenges associated with early detection, accurate classification, and interpretability of diagnostic results in clinical environments.

The project successfully integrates multiple components, including clinical data preprocessing, machine learning-based prediction, MRI-based deep learning classification, and explainable AI mechanisms. Each component plays a crucial role in ensuring accurate and efficient diagnosis of Alzheimer's disease. The use of structured clinical features such as age, BMI, MMSE score, behavioral indicators, and medical history enables effective early-stage prediction, while MRI-based analysis captures structural brain changes associated with disease progression.

One of the key strengths of the system is its ability to provide both diagnostic results and human-readable explanations. The integration of explainable AI enhances transparency and allows healthcare professionals to understand the reasoning behind predictions and classifications. This improves trust, usability, and supports informed clinical decision-making.

While the system demonstrates high accuracy and reliability, it has certain limitations. The performance of the models depends on the quality, size, and diversity of the clinical and MRI datasets. Additionally, variations in imaging quality and patient data may affect prediction outcomes. Therefore, the system should be considered as a decision support tool rather than a replacement for professional medical expertise.

In conclusion, the project highlights the potential of combining machine learning, deep learning, and explainable AI for intelligent healthcare

applications. It provides a strong foundation for future advancements in AI-based medical diagnosis systems. With further improvements such as larger datasets, enhanced model architectures, and integration with real-time healthcare systems, the proposed system can significantly improve the accuracy, efficiency, and reliability of Alzheimer's disease diagnosis.

6.2 FUTURE ENHANCEMENTS

Although the proposed Hybrid Multi-Modal Clinical Decision Support System demonstrates effective performance, there are several areas where improvements can further enhance its accuracy, functionality, and usability in real-world healthcare environments.

One major enhancement is the use of more advanced deep learning architectures. While the current ResNet-18 model provides high classification accuracy, more sophisticated models such as transformer-based architectures or hybrid deep learning frameworks can further improve performance by capturing more complex spatial and feature relationships in MRI images.

Another important improvement is the expansion of datasets. A larger and more diverse collection of clinical data and MRI images from different populations, age groups, and medical conditions would help the system generalize better and improve robustness. This would also reduce bias and enhance the reliability of predictions across different clinical scenarios.

The system can also be enhanced by improving the explainable AI component. More advanced explainability techniques can be integrated to provide deeper insights into feature contributions and decision-making processes. This would further strengthen trust among healthcare professionals and improve the interpretability of results.

Additionally, the system can be extended to support real-time clinical applications. Integration with hospital information systems and electronic health records (EHR) would enable seamless data flow and allow real-time diagnosis and monitoring of patients. This would significantly improve the practical usability of the system in clinical environments.

From a usability perspective, the system can be enhanced by developing more advanced and interactive user interfaces. Features such as dynamic dashboards, detailed analytics, and visualization tools can improve user experience and make the system more accessible to healthcare professionals.

Furthermore, integrating the system with cloud-based platforms and distributed computing frameworks can improve scalability and enable handling of large-scale medical datasets efficiently. This would support deployment in large hospitals and healthcare networks.

In summary, future enhancements should focus on improving model performance, expanding dataset diversity, enhancing explainability, enabling real-time integration, and improving system usability. These improvements can significantly increase the effectiveness and practical applicability of the system in Alzheimer's disease diagnosis and healthcare decision support.

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APPENDIX

SAMPLE SOURCE CODE

```
import torch
import torch.nn as nn
from torchvision import transforms
from torchvision.models import resnet18
from PIL import Image
from prediction.csv_prediction import predict_from_clinical_form
from recommendation.recommendation_engine import get_recommendation
from llm.llm_interface import explain_clinical_decision, explain_mri_stage
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = resnet18(weights=None)
model.fc = nn.Linear(model.fc.in_features, 4)
model.load_state_dict(
    torch.load("models/mri_resnet18_stage.pth", map_location=device,
weights_only=True)
model.eval().to(device)
# Image preprocessing
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize([0.485, 0.456, 0.406],
[0.229, 0.224, 0.225] )])
ORIGINAL_LABELS = {
    0: "Mild_Impairment",
    1: "Moderate_Impairment",
    2: "No_Impairment",
    3: "Very_Mild_Impairment" }
UPDATED_STAGE_MAP = {
    "No_Impairment": "No Impairment",
```

```

"Very_Mild_Impairment": "Mild Impairment",
"Mild_Impairment": "Moderate Impairment",
"Moderate_Impairment": "Severe Impairment" }
def process_clinical_input(age, bmi, mmse, gender, checkbox_inputs):
    gender_numeric = 0 if gender == "Male" else 1
    clinical_data = {
        "Age": int(age),
        "BMI": float(bmi),
        "MMSE": int(mmse),
        "Gender": gender_numeric,
        "GenderText": gender,
        **checkbox_inputs }
    result = predict_from_clinical_form(clinical_data)
    return clinical_data, result
def predict_mri_stage(image_path):
    img = Image.open(image_path).convert("RGB")
    img_tensor = transform(img).unsqueeze(0).to(device)
    with torch.no_grad():
        output = model(img_tensor)
        predicted_class = torch.argmax(output, dim=1).item()
    raw_label = ORIGINAL_LABELS[predicted_class]
    final_stage = UPDATED_STAGE_MAP [raw_label]
    return final_stage
def generate_recommendations(stage):
    result = get_recommendation(stage)
    return result["Recommendations"]
def generate_clinical_explanation(clinical_result, clinical_data):
    return explain_clinical_decision(clinical_result, clinical_data)
def generate_mri_explanation(stage, clinical_data):
    return explain_mri_stage(stage, clinical_data)
def full_pipeline(clinical_inputs, image_path=None)

```

```

clinical_data, clinical_result = process_clinical_input(**clinical_inputs)
diagnosis = clinical_result["Diagnosis"]
if "no" in diagnosis.lower():
    return {
        "Diagnosis": diagnosis,
        "Explanation": generate_clinical_explanation(clinical_result,
clinical_data)
    }
if image_path:
    stage = predict_mri_stage(image_path)
    recommendations = generate_recommendations(stage)
    explanation = generate_mri_explanation(stage, clinical_data)
    return {
        "Diagnosis": diagnosis,
        "Stage": stage,
        "Recommendations": recommendations,
        "Explanation": explanation }
return {
    "Diagnosis": diagnosis,
    "Explanation": generate_clinical_explanation(clinical_result,
clinical_data) }

```


CERTIFICATIONS

TECHNICAL CERTIFICATIONS:

As part of the project development process, relevant certifications were completed to strengthen knowledge in artificial intelligence and machine learning. These certifications provided both theoretical understanding and practical exposure required for designing intelligent, data-driven systems. The acquired skills played a significant role in developing the proposed clinical decision support system for Alzheimer's disease diagnosis, ensuring accuracy, efficiency, and reliability.

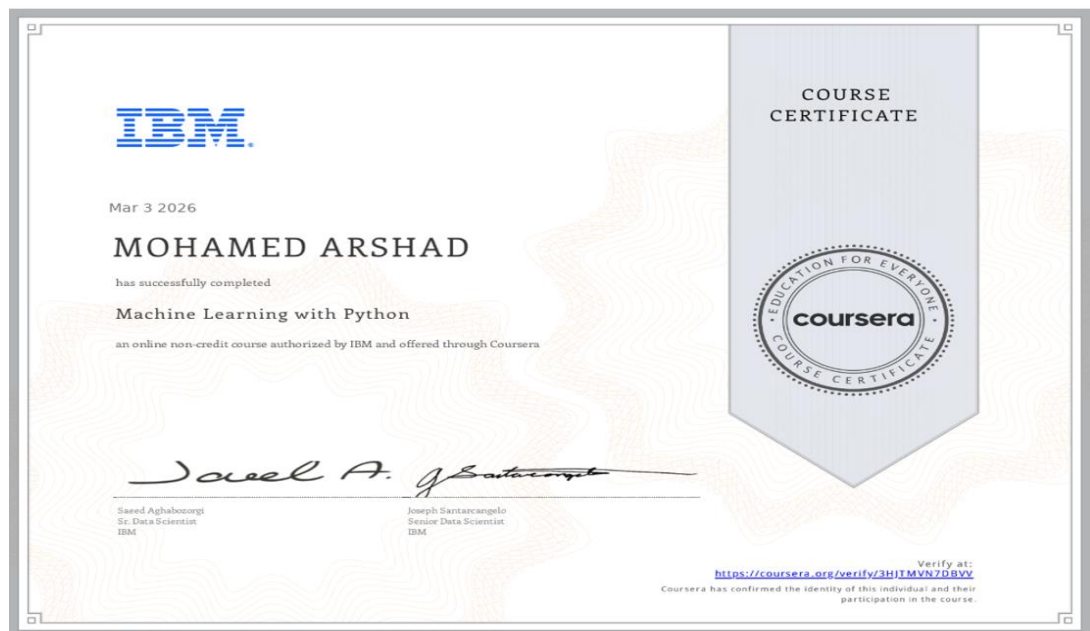
The Oracle Certified AI Foundation *Associate* certification provided a strong foundation in artificial intelligence concepts, including machine learning fundamentals, neural networks, computer vision, and natural language processing. It enhanced understanding of AI-driven system design and helped in integrating intelligent components such as deep learning models and explainable AI into the proposed system, improving overall performance and interpretability.

Below is the certificate awarded upon successful completion of the course:



The Machine Learning with Python certification focused on core concepts such as data preprocessing, supervised and unsupervised learning, model evaluation, and implementation of machine learning algorithms using Python. It provided hands-on experience in building predictive models and analyzing datasets, which directly contributed to the development of the clinical prediction module in the proposed system.

Below is the certificate awarded upon successful completion of the course:



PUBLICATION

JOURNAL PUBLICATION:

The research paper titled **"INTEGRATION OF CLINICAL MACHINE LEARNING, NEUROIMAGING AND LLM FOR ALZHEIMER'S DISEASE DECISION SUPPORT SYSTEM"** was successfully published in the **JETIR** Journal.

The publication highlights the development of a multi-modal system for accurate prediction and classification of Alzheimer's disease using clinical data and MRI images. The system integrates machine learning for clinical prediction, deep learning (ResNet-18) for MRI classification, and LLM for generating human-readable insights, providing an efficient diagnostic solution. The publication validates the effectiveness and technical strength of the proposed system in AI-based healthcare applications.

Below is the certificate awarded for the successful publication of the research paper:



Mohamed Arshad J and Dr. Syed Masood M, "Integration Of Clinical Machine Learning, Neuroimaging And Llm For Alzheimer's Disease Decision Support System," JETIR Journal of Emerging Technologies and Innovative Research, ISSN 2349-5162 UGC Approved, Published in Volume 13 Issue 4 April-2026

TECHNICAL BIOGRAPHY



MOHAMED ARSHAD J (RRN: 240282601050), was born on 08th JAN 2003. He has completed his Bachelor of Computer Application Specialized in Cloud Technology and Information Security in B.S Abdur Rahman Crescent Institute of Science and Technology. Currently He is pursuing a Master of Computer Application in B.S Abdur Rahman Crescent Institute of Science and Technology, Vandalur, Chennai 600048.

He developed a major project titled *“Integration of Clinical Machine Learning Neuroimaging and LLM for Alzheimer’s Disease Decision Support System,”* aimed at early detection and diagnosis of Alzheimer’s disease using clinical data and MRI images to enhance healthcare decision-making. He has completed certifications in *Machine Learning with Python* and *Oracle Certified AI Foundation Associate*, and published a research paper in *JETIR*, reflecting his academic involvement. Skilled in Python, Scikit-learn, and PyTorch, with experience in deep learning models like ResNet-18, he is proficient in data preprocessing, model evaluation, and explainable AI. His career objective is to contribute to innovative AI-based healthcare systems that improve diagnostic accuracy and clinical decision support.

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